

Single nuclei transcriptomics reveal the differentiation trajectories of periosteal skeletal/stem progenitor cells in bone regeneration

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This **fundamental** study generated a single cell atlas of mouse periosteal cells under both steady-state and fracture healing conditions to address the knowledge gap regarding cellular composition of the periosteum and their responses to injury. Based on **convincing** transcriptome analyses and experimental validation, the authors identified the injury induced fibrogenic cell (IIFC) as a characteristic cell type appearing in the bone regeneration process and proposed that the IIFC is a progenitor undergoing osteochondrogenic differentiation. This study will provide a significant publicly accessible dataset to reexamine the expression of the reported periosteal stem and progenitor cell markers.

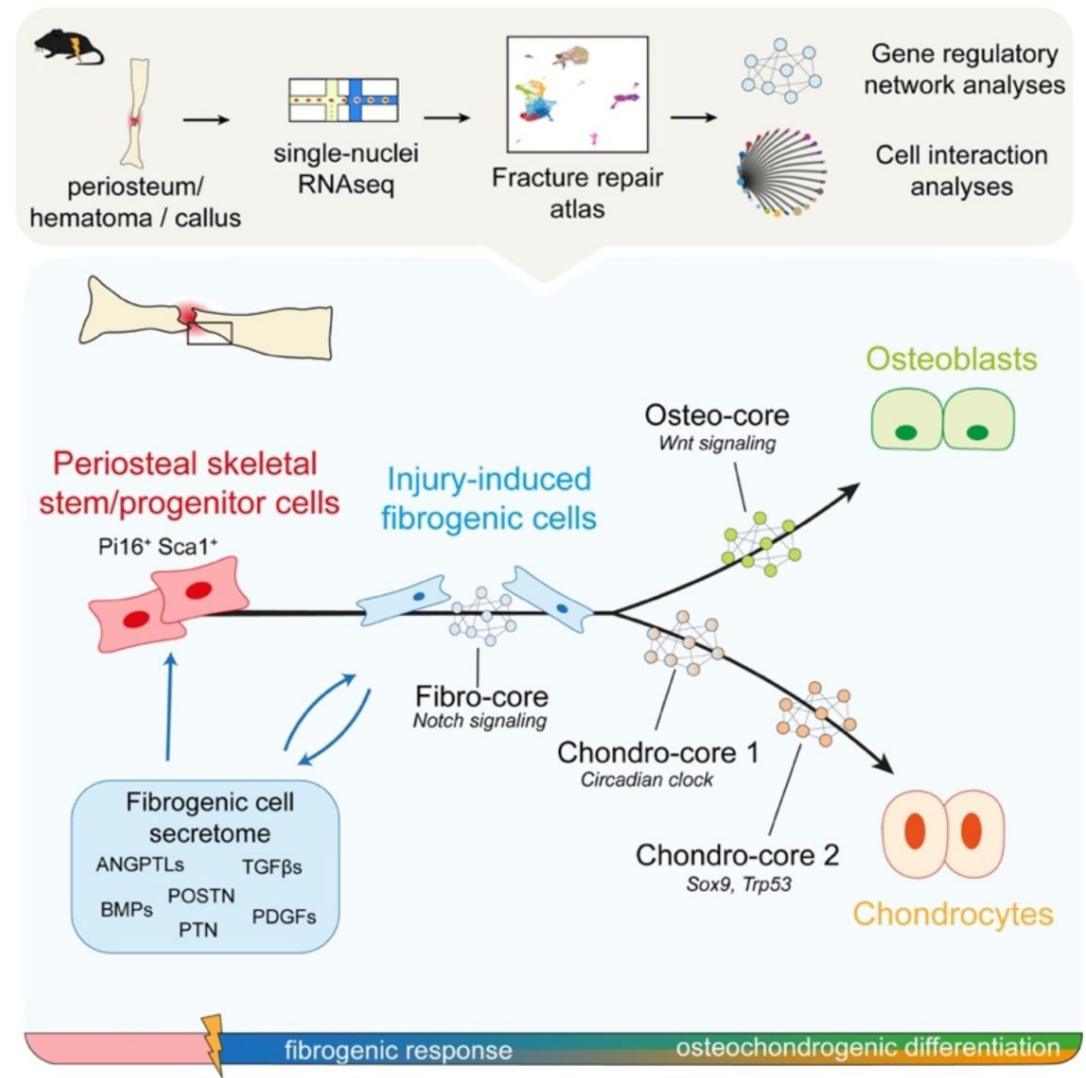
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Abstract

Bone regeneration is mediated by skeletal stem/progenitor cells (SSPCs) that are mainly recruited from the periosteum after bone injury. The composition of the periosteum and the steps of SSPC activation and differentiation remain poorly understood. Here, we generated a single-nuclei atlas of the periosteum at steady-state and of the fracture site during early stages of bone repair (<https://fracture-repair-atlas.cells.ucsc.edu>). We identified periosteal SSPCs expressing stemness markers (*Pi16* and *Ly6a/SCA1*) and responding to fracture by adopting an injury-induced fibrogenic cell (IIFC) fate, prior to undergoing osteogenesis or chondrogenesis. We identified distinct gene cores associated with IIFCs and their engagement into osteogenesis and chondrogenesis involving Notch, Wnt and the circadian clock signaling respectively. Finally, we show that IIFCs are the main source of paracrine signals in the fracture environment, suggesting a crucial paracrine role of this transient IIFC population during fracture healing. Overall, our study provides a complete temporal topography of the

early stages of fracture healing and the dynamic response of periosteal SSPCs to injury, redefining our knowledge of bone regeneration.

Graphical Abstract



Introduction

The skeleton provides structural support and protection for internal organs in the vertebrate body. Bones can fracture but regenerate themselves efficiently without scarring. Bone regeneration is mediated by the action of resident skeletal stem/progenitor cells (SSPCs) from periosteum and bone marrow, and SSPCs from skeletal muscle adjacent to the fracture site^{1,2,3,4}. SSPCs are activated during the inflammatory phase of healing and differentiate into osteoblasts and/or chondrocytes to repair bone via a combination of intramembranous and endochondral ossification. During intramembranous ossification SSPCs differentiate directly into osteoblasts, while during endochondral ossification SSPCs first differentiate into chondrocytes to form an intermediate cartilage template subsequently replaced by bone.

The periosteum, an heterogeneous tissue located on the outer surface of bones, is a crucial source of SSPCs during bone healing^{1,2,5,6-7}. Periosteal SSPCs exhibit a high regenerative potential. They also display both osteogenic and chondrogenic potentials after injury compared to bone marrow SSPCs that are mostly osteogenic and skeletal muscle SSPCs that are mostly chondrogenic^{1,4,8}. Periosteal SSPCs (pSSPCs) are still poorly characterized and their response to bone injury remain elusive. Recent advances in single cell/nuclei transcriptomic analyses provided new insights in stem cell population heterogeneity and regeneration processes in many organs⁹⁻¹¹. However, few studies have investigated bone fracture healing at the single cell level and these studies have focused on cultured cells or late stages of bone repair^{3,12}. Therefore, a complete dataset of periosteum and bone regeneration is lacking and is essential to decipher the mechanisms of pSSPC activation and regulation. Here, we created a single-nuclei atlas of the uninjured periosteum and its response to bone fracture. We generated single-nuclei RNAseq (snRNAseq) datasets from the uninjured periosteum and from the periosteum and hematoma/callus at days 3, 5 and 7 post-tibial fracture. Our analyses thoroughly describe the heterogeneity of the periosteum at steady state and the steps of pSSPC activation and differentiation after injury. We show that pSSPCs represent a single population localized in the fibrous layer of the periosteum. Periosteal SSPCs can provide osteoblasts and chondrocytes for bone repair by first generating a common injury-induced fibrogenic cell (IIFC) population that can then engage into osteogenesis and chondrogenesis. We identified the gene networks regulating pSSPC fate after injury and IIFCs as the main population producing paracrine factors mediating the initiation of bone healing.

Results

Heterogeneity of the periosteum at steady state

To investigate the heterogeneity of the periosteum at steady-state, we performed snRNAseq of the periosteum of wild-type mice (**Fig. 1A-B**). Single nuclei transcriptomics was previously shown to provide results equivalent to single cell transcriptomics, but with better cell type representation and reduced digestion-induced stress response¹³⁻¹⁶. After filtering, we obtained 1189 nuclei, corresponding to 8 cell populations: SSPC (expressing *Pi16*), fibroblasts (expressing *Pdgfra*), osteogenic cells (expressing *Runx2*), Schwann cells (expressing *Mpz*), pericytes/SMC (expressing *Tagln*), immune cells (expressing *Ptprc*), adipocytes (expressing *Lpl*), and endothelial cells (ECs, expressing *Pecam1*) (**Fig. 1C-D**, **Fig. 1 – Supplementary Figure 1A**). We performed in depth analyses of the SSPC, fibroblast and osteogenic cell populations. Subset analyses of clusters 0 to 5 identified 5 distinct SSPC/fibroblast populations expressing *Pdgfra* and *Prrx1*: *Pi16*⁺ *Ly6a* (SCA1)⁺ cells (cluster 0), *Csmd1*⁺ cells (cluster 1), *Hsd11b1*⁺ cells (cluster 2), *Cldn1*⁺ cells (cluster 3), and *Luzp2*⁺ cells (cluster 4) (**Fig. 2A-B**). Cluster 0 is the only cell cluster containing cells expressing *Pi16*⁺ and stemness markers including *Ly6a* (SCA1), *Dpp4*, and *Cd34* (**Fig. 2C-D**). Cytotrace scoring identified *Pi16*⁺ cells as the population in the most undifferentiated state (**Fig. 2E**). We performed in vitro CFU assays with sorted GFP⁺SCA1⁺ and GFP⁺SCA1⁻ cells isolated from the periosteum of *Prx1*^{Cre}; *R26*^{mTmG} mice, as *Prx1* labels all SSPCs contributing to the callus formation including *Pi16*⁺ cells (**Fig. 2D**). ¹, *Prx1*-GFP⁺ SCA1⁺ cells showed higher CFU potential compared to GFP⁺SCA1⁻ cells, confirming their stem/progenitor property (**Fig. 2F-G**). Then, we grafted *Prx1*-GFP⁺ SCA1⁺ et *Prx1*-GFP⁺ SCA1⁻ periosteal cells at the fracture site of wild-type mice. Only GFP⁺SCA1⁺ cells formed cartilage after fracture indicating that GFP⁺SCA1⁺ cells encompass periosteal SSPCs with osteochondrogenic potential (**Fig. 2H**). We explored the expression of other known markers of periosteal SSPCs, including *Ctsk*, *Acta2* (αSMA), *Gli1*, and *Mx1*, but no marker was fully specific to one cell cluster (**Fig. 2 – Supplementary Fig. 1**) ^{1,2,5,6-7,17-21}.

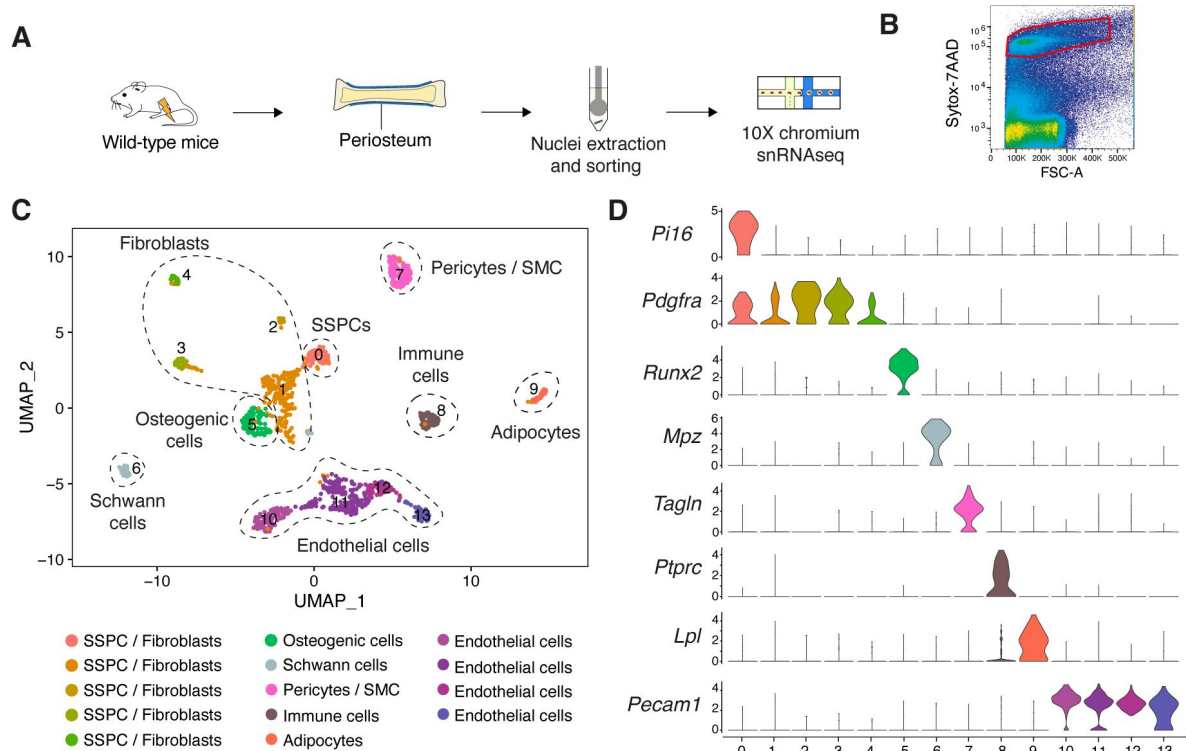


Figure 1

Heterogeneity of the periosteum at steady state

A. Experimental design. Nuclei were extracted from the periosteum of uninjured tibia and processed for single-nuclei RNAseq. **B.** Sorting strategy of nuclei stained with Sytox-7AAD for snRNAseq. Sorted nuclei are delimited by a red box. **C.** UMAP projection of color-coded clustering of the uninjured periosteum dataset. Six populations are identified and delimited by black dashed lines. **D.** Violin plots of key marker genes of the different cell populations.

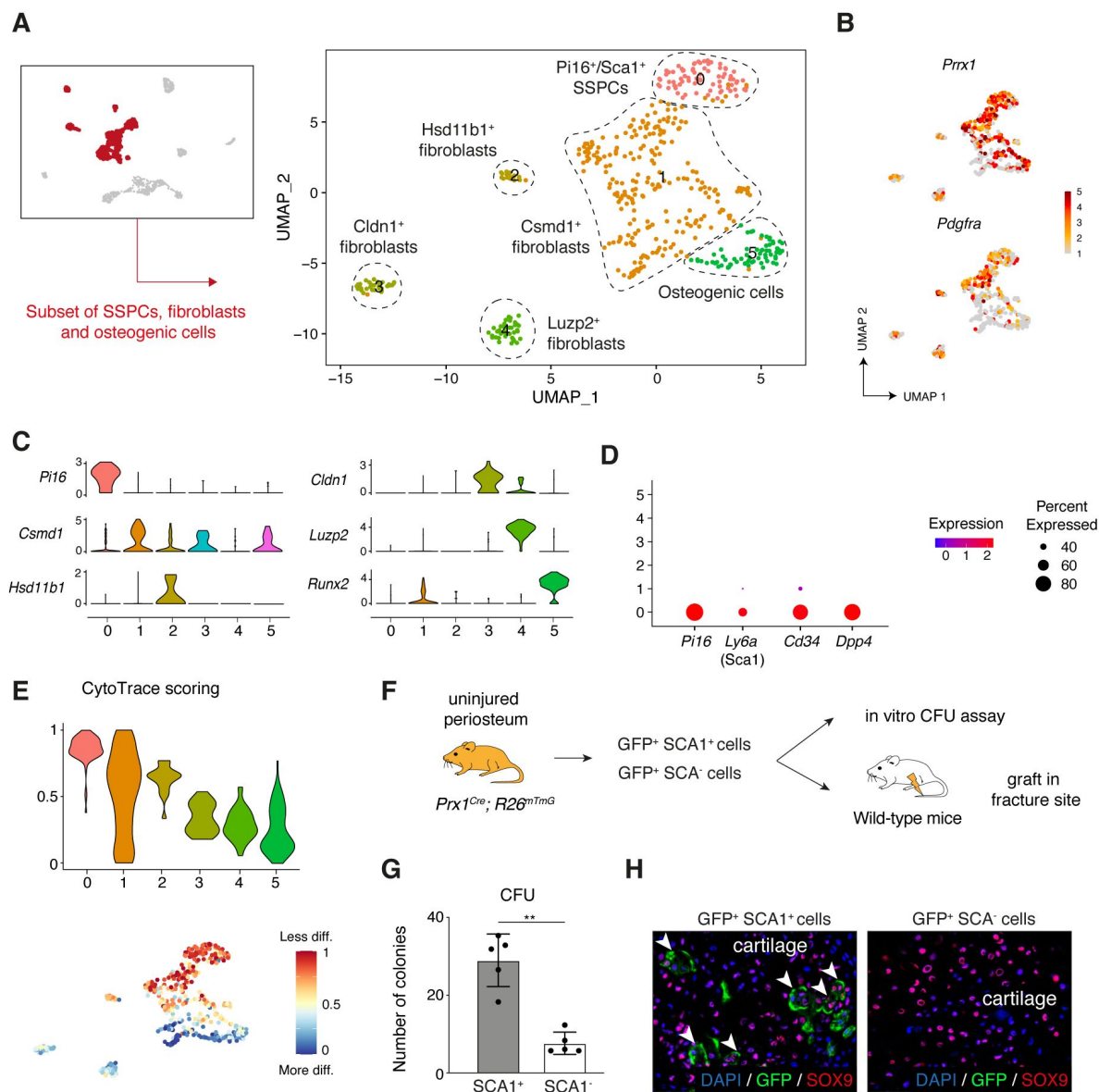


Figure 2

Identification of periosteal SSPCs in the intact periosteum

A. UMAP projection of color-coded clustering of the subset of SSPC/fibroblasts. **B.** Feature plots of *Prx1* and *Pdgfra* in the subset of SSPC/fibroblasts. **C.** Feature plots of key marker genes of the different cell populations. **D.** Dot plot of the stemness markers *Pi16*, *Ly6a* (Sca1), *Cd34*, and *Dpp4*. **E.** Violin and feature plots of CytoTrace scoring in the subset of SSPC/fibroblasts, showing that SCA1⁺ SSPCs (cluster 0) are the less differentiated cells in the dataset. **F.** Experimental design: GFP⁺ SCA1⁺ and GFP⁺ SCA1⁻ were isolated from uninjured tibia of *Prx1^{Cre}; R26^{mTmG}* mice and used for in vitro CFU assays or grafted at the fracture site of wild-type mice. **G.** In vitro CFU assay of murine periosteal *Prx1*-GFP⁺ SCA1⁺ and *Prx1*-GFP⁺ SCA1⁻ cells. **H.** High magnification of SOX9 immunofluorescence of callus section 14 days post-fracture showing that GFP⁺ SCA1⁺ cells contribute to the callus (white arrowheads) while GFP⁺ SCA1⁻ cells are not contributing (n=3 per group).

The fracture repair atlas

To investigate the periosteal response to bone fracture, we collected injured periosteum with hematoma or callus at days 3, 5 and 7 post-fracture, extracted the nuclei and processed them for snRNAseq (**Fig. 3A**). We combined the datasets with the uninjured periosteum from **Figure 1** and obtained a total of 6213 nuclei after filtering. The combined dataset was composed of 25 clusters corresponding to 11 cell populations: SSPCs (expressing *Pi16*), injury-induced fibrogenic cells (IIFCs, expressing ECM-related genes including *Postn*), osteoblasts (expressing *Ibsp*), chondrocytes (expressing *Col2a1*), osteoclasts (expressing *Ctsk*), immune cells (expressing *Ptprc*), Schwann cells (expressing *Mpz*), endothelial cells (expressing *Pecam1*), pericytes (expressing *Rgs5*), SMCs (expressing *Tagln*) and adipocytes (expressing *Lpl*) (**Fig. 3B-C**, **Fig 3 – Supplementary Fig. 1-2**). Next, we observed the dynamics of the cell populations in response to bone fracture (**Fig. 3D**, **Fig 3 – Supplementary Fig. 1B**). After injury, the percentage of SSPCs was strongly decreased and the percentage of IIFCs progressively increased (**Fig. 3D-E**). The percentage of chondrocytes and osteoblasts increased from day 3 post-fracture. Immune cells were drastically increased at day 3 after injury, before progressively decreasing at days 5 and 7 post-fracture (**Fig. 3E**).

Spatial organization of the fracture callus

To evaluate the spatial distribution of the main cell populations identified in the snRNAseq datasets, we performed in situ immunofluorescence and RNAscope experiments on uninjured periosteum and days 3, 5 and 7 post-fracture hematoma/callus tissues. In the uninjured periosteum, we detected *Pi16*-expressing SSPCs, *Postn*-expressing cells, OSX⁺ osteoblasts, CD31⁺ endothelial cells and CD45⁺ immune cells (**Fig 4A-B**). *Pi16*-expressing SSPCs were located within the fibrous layer, while *Postn*-expressing cells were found in the cambium layer and corresponded to *Runx2*-expressing osteogenic cells (**Fig. 4 – Supplementary Fig. 1A-C**). Although *Postn* expression was weak in uninjured periosteum, *Postn* expression was strongly increased in response to fracture, specifically in IIFCs (**Fig. 4 – Supplementary Fig. 1D-E**). At day 3 post-fracture, we observed periosteal thickening and the formation of a fibrous hematoma (**Fig. 4C**). We did not detect *Pi16*-expressing SSPCs, consistent with the absence of cells expressing SSPC markers in the day 3 snRNAseq dataset compared to uninjured periosteum (**Fig. 4 – Supplementary Figure 2**). POSTN⁺ IIFCs and immune cells were the main populations present in hematoma and activated periosteum. Few IIFCs in the activated periosteum expressed SOX9 and OSX and only the periosteum was vascularized (**Fig. 4D**). At days 5 and 7 post-fracture, the callus was formed mainly of fibrotic tissue, new bone formed on the periosteal surface at the periphery of the callus and small cartilage islets were detected in the center of the callus near the periosteal surface (**Fig. 4E-H**). The fibrotic tissue contained mostly IIFCs, as well as immune and endothelial cells. We also observed SOX9⁺ and OSX⁺ cells in the fibrotic tissue, while the cartilage was solely composed of SOX9⁺ chondrocytes. OSX⁺ osteoblasts were the main cell population detected in the new bone at the periosteal surface. We also observed a progressive reduction of POSTN⁺ cells and immune cells from day 5 to 7 and increased vascularization in the newly formed bone (**Fig. 4E-H**).

Periosteal SSPCs differentiate via an injury-induced fibrogenic stage

To understand the differentiation and fate of pSSPCs after fracture, we analyzed the subset of SSPC, IIFC, chondrocyte and osteoblast clusters from the combined fracture dataset (**Fig. 5A**, **Fig. 5 – Supplementary Figure 1**). We performed pseudotime analyses to determine the differentiation trajectories, defining the starting point in the pSSPC population, where the uninjured and undifferentiated cells cluster. We identified that pSSPCs differentiate in 3 stages

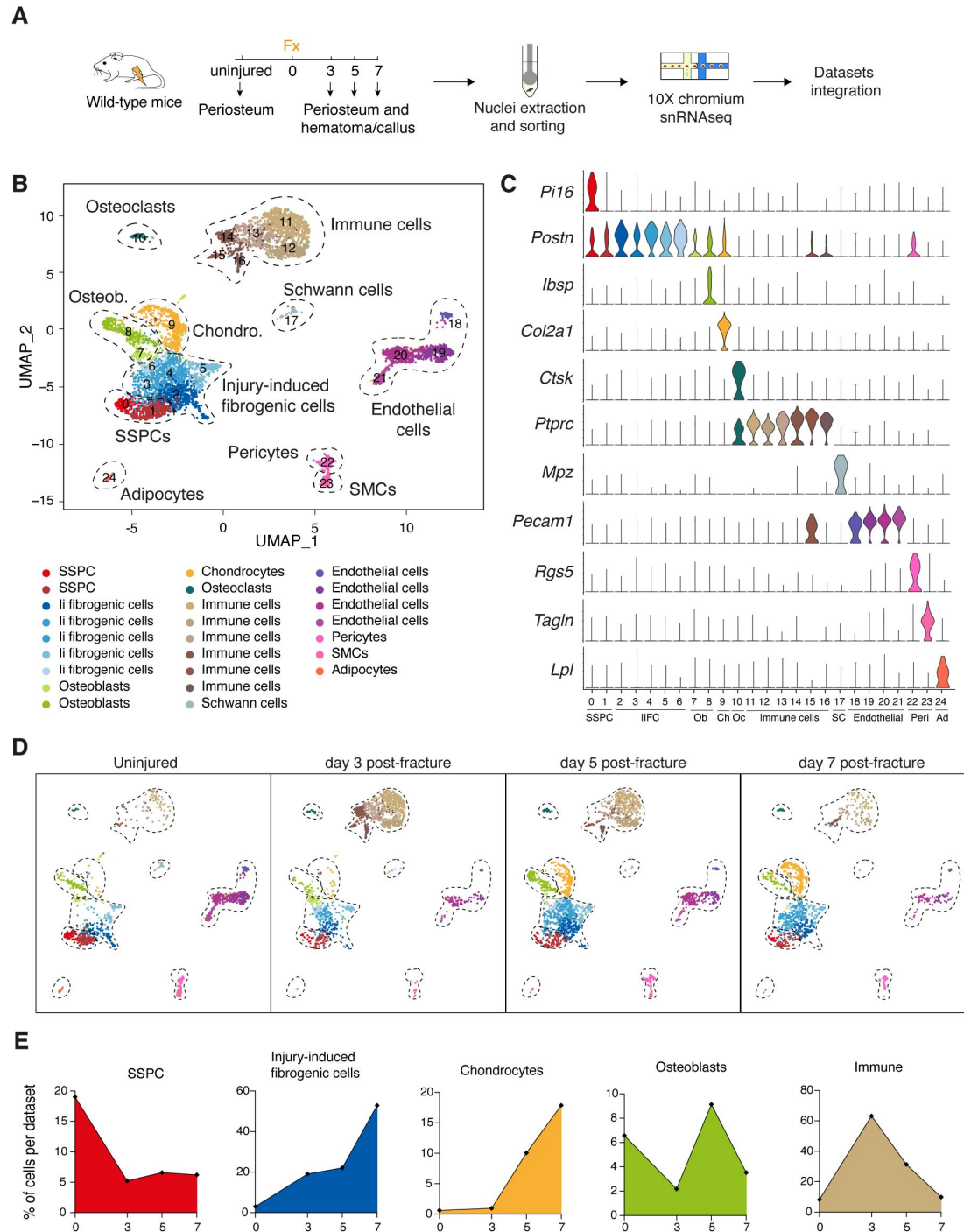


Figure 3

Periosteal response to fracture at single-nuclei resolution

A. Experimental design. Nuclei were extracted from the periosteum of uninjured tibia of wild-type mice and from the injured periosteum and hematoma/callus at days 3, 5 and 7 post-tibial fracture and processed for single-nuclei RNAseq. **B.** UMAP projection of color-coded clustering of the integration of uninjured, day 3, day 5 and day 7 datasets. Eleven populations are identified and delimited by black dashed lines. **C.** Violin plots of key marker genes of the different cell populations. **D.** UMAP projection of the combined dataset separated by time-point. **E.** Percentage of cells in SSPC, injury-induced fibrogenic cell, osteoblast, chondrocyte, and immune cell clusters in uninjured, day 3, day 5 and day 7 datasets.

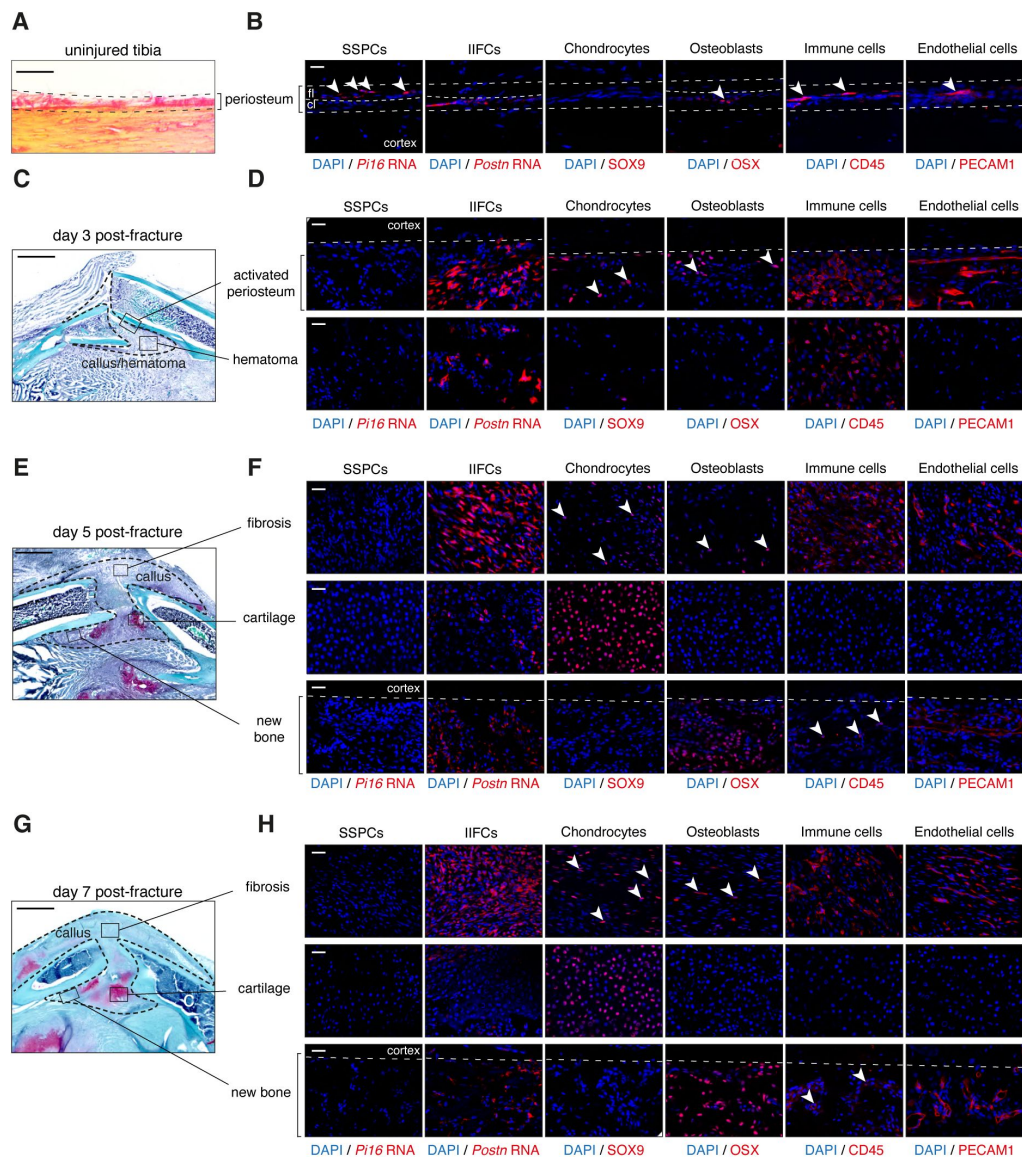


Figure 4

Cellular organization of the fracture callus

A. Picrosirius staining of the uninjured periosteum. **B.** Immunofluorescence and RNAscope on adjacent sections show the presence of SSPCs (*Pi16*-expressing cells) in the fibrous layer (fl), *Postn*-expressing cells in the cambium layer (cl), *OSX*⁺ osteoblasts, immune cells (*CD45*⁺) and endothelial cells (*PECAM1*⁺) in the periosteum (n=3 per group). **C.** Safranin'O staining of longitudinal callus sections at day 3-post tibial fracture. **D.** Immunofluorescence and RNAscope on adjacent sections show absence of SSPCs (*Pi16*⁺), and presence of IIFCs (*Postn*⁺) and immune cells (*CD45*⁺) in the activated periosteum and hematoma at day 3-post fracture. Chondrocytes (*SOX9*⁺, white arrowhead), osteoblasts (*OSX*⁺, white arrowhead), immune cells (*CD45*⁺) and endothelial cells (*PECAM1*⁺) are detected in the activated periosteum (n=3 per group). **E.** Safranin'O staining of longitudinal callus sections at day 5-post tibial fracture. **F.** Immunofluorescence and RNAscope on adjacent sections show IIFCs (*Postn*⁺), chondrocytes (*SOX9*⁺, white arrowhead), osteoblasts (*OSX*⁺, white arrowhead), immune cells (*CD45*⁺) and endothelial cells (*PECAM1*⁺) in the fibrosis, chondrocytes (*SOX9*⁺) in the cartilage and osteoblasts (*OSX*⁺), immune cells (*CD45*⁺, white arrowhead) and endothelial cells (*PECAM1*⁺) in the new bone. (n=3 per group). **G.** Safranin'O staining of longitudinal callus sections at day 7-post tibial fracture. **H.** Immunofluorescence and RNAscope on adjacent sections show IIFCs (*Postn*⁺), chondrocytes (*SOX9*⁺, white arrowhead), osteoblasts (*OSX*⁺, white arrowhead), immune cells (*CD45*⁺) and endothelial cells (*PECAM1*⁺) in the fibrosis, chondrocytes (*SOX9*⁺) in the cartilage and osteoblasts (*OSX*⁺), immune cells (*CD45*⁺, white arrowhead) and endothelial cells (*PECAM1*⁺) in the new bone (n=3 per group). Scale bars: A-B-E: 1mm, B-D-F: 100µm.

starting from the pSSPC population (expressing *Ly6a*, *Pi16* and *Cd34*), predominant in uninjured dataset (Fig. 5B–C). Periosteal SSPCs then transition through an injury-induced fibrogenic stage predominant at days 3 and 5 post-injury. In this intermediate fibrogenic stage, IIFCs express high levels of extracellular matrix genes, such as *Postn*, *Aspn* and collagens. Subsequently, IIFCs differentiate into chondrocytes (expressing *Acan*, *Col2a1* and *Sox9*) or osteoblasts (expressing *Sp7*, *Alpl* and *Ibsp*), both predominant at days 5 and 7 (Fig. 5B–C, Table S1). We observed a parallel between pseudotime and the timepoints of the dataset, confirming that the differentiation trajectory follows the timing of cell differentiation (Fig. 5D). These results show that pSSPCs respond to fracture via an injury-induced fibrogenic stage common to chondrogenesis and osteogenesis and independent of their final fate (Fig. 5E). To visualize the transition of IIFCs towards chondrocytes and osteoblasts in the fracture callus, we performed co-immunofluorescence on day 5 post-fracture hematoma/callus. We observed a progressive increase in SOX9 and OSX signals in IIFCs at the fibrosis-to-cartilage and fibrosis-to-bone transition zones respectively (Fig. 6A). To functionally validate the steps of pSSPC activation, we isolated SCA1⁺ GFP⁺ pSSPCs from *Prx1*^{Cre}; *R26*^{mTmG} mice, excluding endothelial cells (SCA1+GFP⁻) and pericytes (SCA1-GFP⁺), and grafted them at the fracture site of wild-type hosts (Fig. 6B, Figure 6 – Supplementary Figure 1). We observed that grafted GFP⁺ pSSPCs formed POSTN⁺ IIFCs at day 5 post-fracture (Fig. 6B). Then, we isolated IIFCs, that correspond to GFP⁺ CD146⁻ cells from the day 3 post-fracture callus of *Prx1*^{Cre}; *R26*^{mTmG} mice without contamination by pericytes (GFP⁺CD146⁺ cells) (Fig. 6C, Figure 6 – Supplementary Figure 1). We grafted the GFP⁺ IIFCs at the fracture site of wild-type hosts and showed that grafted cells formed bone and cartilage at day 14 post-fracture. These results confirmed that pSSPCs first become IIFCs that differentiate into osteoblasts and chondrocytes.

Characterization of injury-induced fibrogenic cells

We performed in depth analyses of the newly identified IIFC population. Gene ontology (GO) analyses of upregulated genes in IIFCs (clusters 2 to 6) showed enrichment in GOs related to tissue development, extracellular matrix (ECM), and ossification (Fig. 7A). We identified several ECM-related genes specifically upregulated in IIFCs, including collagens (*Col3a1*, *Col5a1*, *Col8a1*, *Col12a1*), *Postn*, and *Aspn* (Fig. 7B–C, Fig. 7 – Supplementary Figure 1). We also identified GO terms related to cell signaling, migration, differentiation, and proliferation, indicating that this step corresponds to an injury response. Only a small subset of IIFCs undergo apoptosis, further supporting that IIFCs are maintained in the fracture environment giving rise to osteoblasts and chondrocytes (Fig. 7 – Supplementary Figure 2). To further understand the mechanisms regulating SSPC activation and fate after injury, we performed gene regulatory network (GRN) analyses on the subset of SSPCs, IIFCs, osteoblasts and chondrocytes using SCENIC package (Single Cell rEgulatory Network Inference and Clustering)²². We identified 280 activated regulons (transcription factor/TF and their target genes) in the subset dataset. We performed GRN-based tSNE clustering and identified SSPC, IIFC, chondrocyte and osteoblast populations (Fig. 7D–E, Fig. 7-Supplementary Figure 3A). Fibroblasts from uninjured periosteum (*Hsd11b1*⁺, *Cldn1*⁺ and *Luzp2*⁺ cells and corresponding to cluster 10 of Fig. 5B) clustered separately from the other populations, suggesting the absence of their contribution to bone healing. Analysis of the number of activated regulons per cell indicated that SSPCs are the most stable cell population (higher number of activated regulons), while IIFCs are the less stable population, confirming their transient state (Fig. 7F). We then investigated cell population specific regulons. SSPCs showed activated regulons linked to stemness, including *Hoxa10* (16g), *Klf4* (346g), *Pitx1* (10g), and *Mta3* (228g)^{23–25}, and immune response, including *Stat6* (30g), *Fiz1* (72g) and *Stat5b* (18g)^{26–28} (Fig. 7G). Osteoblasts and chondrocytes display cell specific activated regulons including *Sp7* (18g) and *Sox9* (77g) respectively. We identified 21 regulons that we named fibro-core and that were upregulated specifically in the IIFC population (Fig. 7H–I, Table S2). Several fibro-core regulons, such as *Meis1* (1556g), *Pbx1* (11g), *Six1* (20g), and *Pbx3* (188g) are known to be involved in cell differentiation during tissue development and repair^{29–32}. Reactome pathway analysis

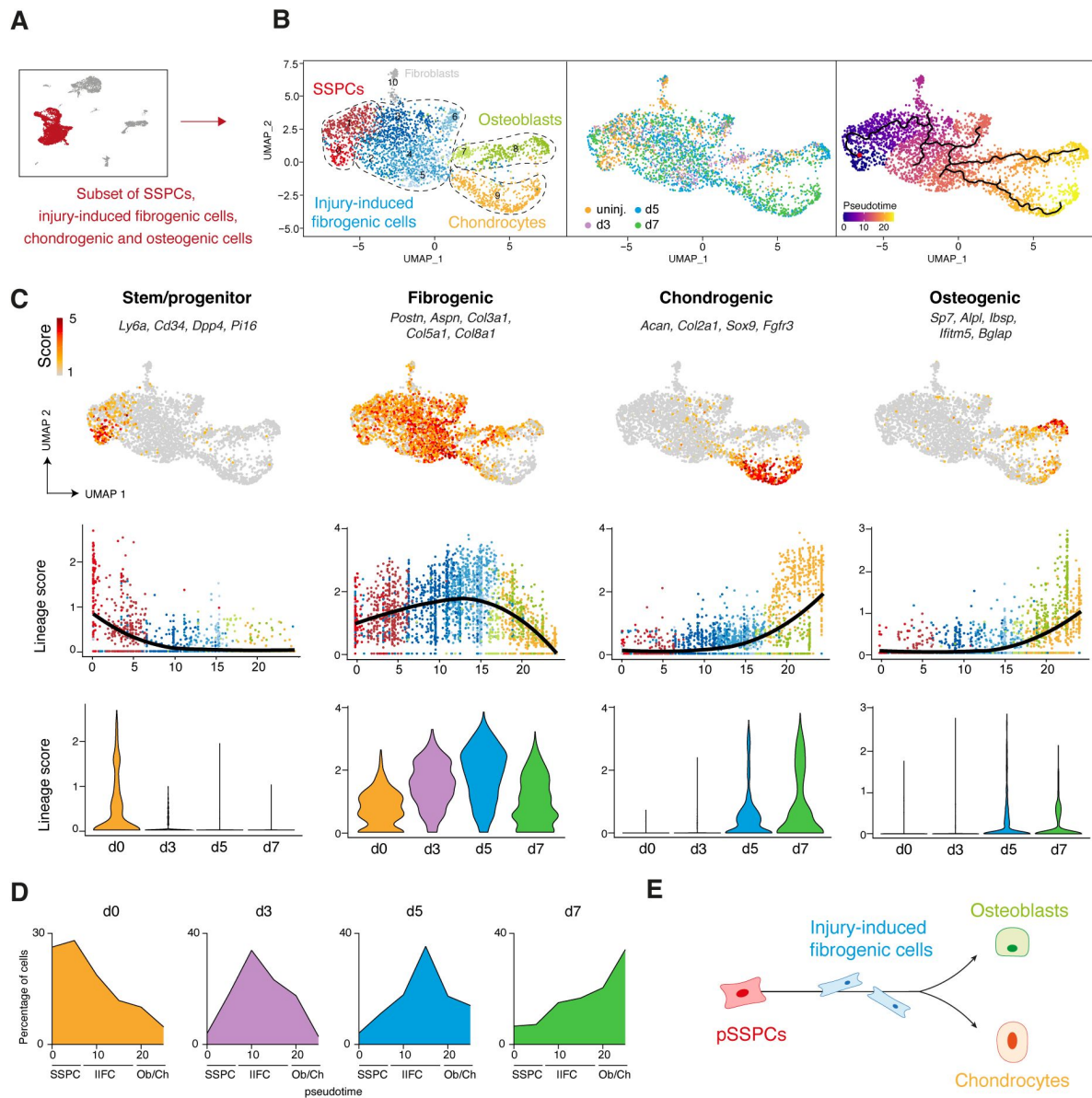


Figure 5

Periosteal SSPCs activate through a common fibrogenic state prior to undergoing osteogenesis or chondrogenesis

A. SSPCs, injury-induced fibrogenic cells (IIFCs), chondrocytes and osteoblasts from integrated uninjured, day 3, day 5 and day 7 post-fracture samples were extracted for a subset analysis. **B.** UMAP projection of color-coded clustering (left), color-coded sampling (middle) and monocle pseudotime trajectory (right) of the subset dataset. The four populations are delimited by black dashed lines. **C.** (top) Feature plots of the stem/progenitor, fibrogenic, chondrogenic and osteogenic lineage scores (middle) Scatter plot of the lineage scores along pseudotime. (bottom) Violin plot of the lineage score per time point. **D.** Distribution of the cells along the pseudotime per timepoint. **E.** Schematic representation of the activation trajectory of pSSPCs after fracture.

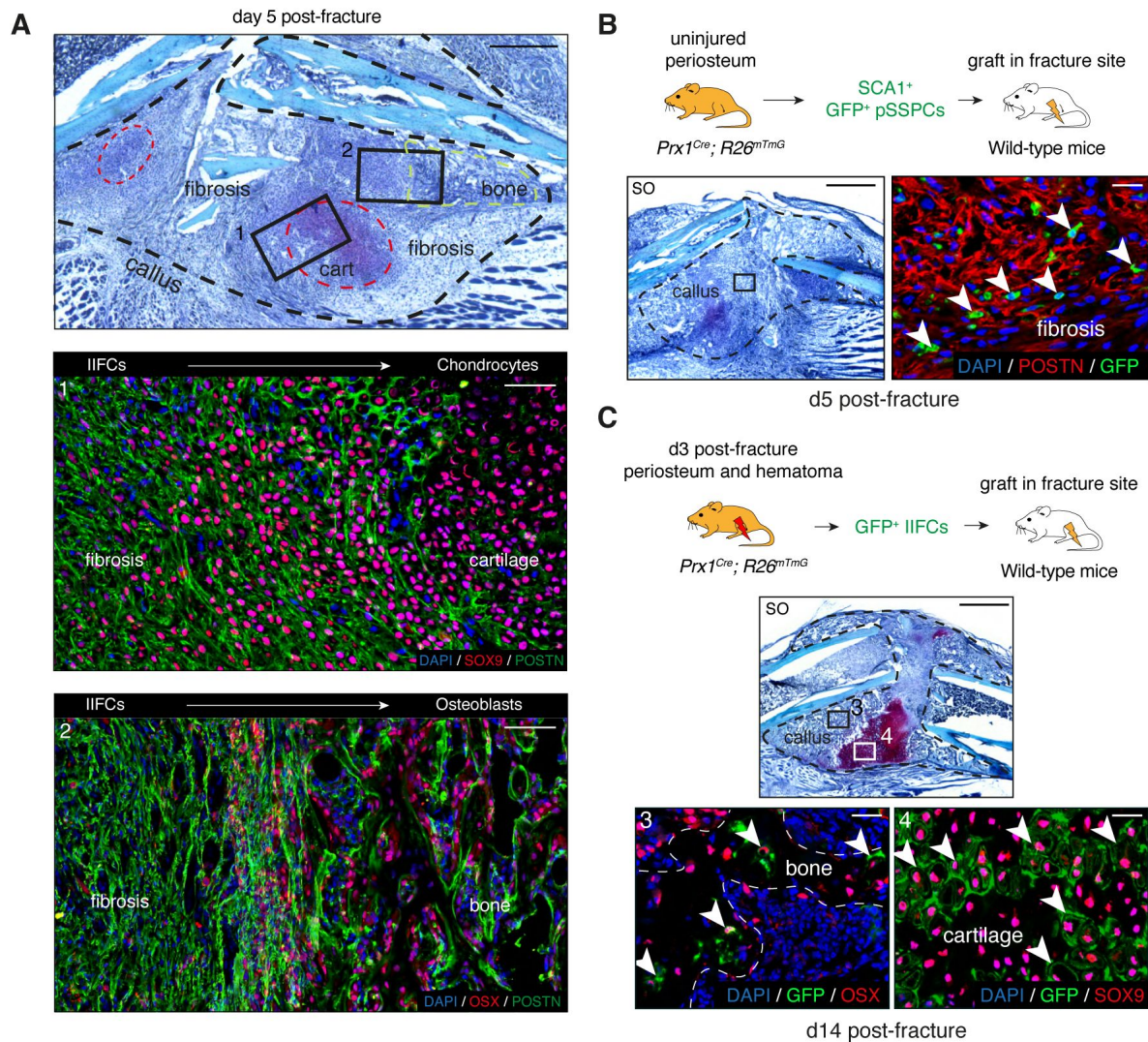


Figure 6

In vivo validation of pSSPC activation trajectory

A. (Top) Representative Safranin'O staining on longitudinal sections of the hematoma/callus at day 5 post-fracture. The callus is composed of fibrosis, cartilage (red dashed line) and bone (green dashed line). (Middle, box 1) Immunofluorescence on adjacent section shows decreased expression of POSTN (green) and increased expression of SOX9 (red) in the fibrosis-to-cartilage transition zone. (Bottom, box 2) Immunofluorescence on adjacent section shows decreased expression of POSTN (green) and increased expression of OSX (red) in the fibrosis-to-bone transition zone (n=3 per group). **B.** Experimental design: GFP⁺ SCA1⁺ SSPCs were isolated from uninjured tibia of *Prx1^{Cre}; R26^{mTmG}* mice and grafted at the fracture site of wild-type mice. Safranin'O staining of callus sections at day 5 post-fracture and high magnification of POSTN immunofluorescence of adjacent section showing that GFP⁺ cells contribute to the callus and that grafted SSPCs differentiate into POSTN⁺ IIFCS (white arrowheads) (n=4 per group). **C.** Experimental design: GFP⁺ IIFCs from periosteum and hematoma at day 3 post-fracture tibia were isolated from *Prx1^{Cre}; R26^{mTmG}* mice and grafted at the fracture site of wild-type mice. Safranin'O of callus sections at day 14 post-fracture and high magnification of OSX and SOX9 immunofluorescence of adjacent sections showing that GFP⁺ cells contribute to the callus and that grafted IIFCs differentiate into OSX⁺ osteoblasts (box 3, white arrowheads) and SOX9⁺ chondrocytes (box 4, white arrowheads) (n=4 per group). Scale bars: Low magnification: A: 500µm, B-C: 1mm. High magnification: 100µm.

showed that the most significant terms linked to IIFC-specific TFs are related to Notch signaling (Fig. 7J [↗](#), Table S3 [↗](#)). We confirmed that Notch signaling is increased in the IIFC stage (Fig. 7J [↗](#)), suggesting its involvement in the fibrogenic phase of bone repair.

Distinct gene cores regulate the engagement of IIFCs in chondrogenesis and osteogenesis

We sought to identify the drivers of the transition of IIFCs to chondrocytes or osteoblasts. We identified 2 cores of regulons involved in chondrogenic differentiation. Chondro-core 1 is composed of 9 regulons specific to the transition of IIFCs to chondrocytes, including *Maf* (17g), *Arntl* (1198g), and *Nfatc2* (37g) and chondro-core 2, composed of 14 regulons specific to differentiated chondrocytes (Fig. 8A [↗](#), Fig. 8-Supplementary Figure 1A [↗](#)). Chondro-core 1 regulons are known to be regulators of the circadian clock (*Npas2*, *Arntl*) or of the T/B-cell receptor cellular response (*Bach2*, *Nfact2*, *Nfkb1*). STRING network analysis showed that TFs from the chondro-core 1 are interacting between each other and are at the center of the interactions between the chondro-core 2 TFs, including *Sox9*, *Trp53* and *Mef2c* (Fig. 8B [↗](#), Fig. 8-Supplementary Figure 1B [↗](#)). We observed that chondro-core 1 is only transiently activated when IIFCs are engaging into chondrogenesis and precedes the activation of chondro-core 2 (Fig. 8C-D [↗](#)). Chondro-core 1 activity was high in early differentiated chondrocytes (low *Acan* expression) and progressively reduced as chondrocytes undergo differentiation, while chondro-core 2 activity was gradually increased as chondrocytes differentiate (Fig. 8D [↗](#)). This suggests that transient activation of the chondro-core 1 allows the transition of IIFCs into chondrocytes. Then, we investigated the osteogenic commitment of IIFCs. We identified 8 regulons forming the osteo-core and activated in IIFCs transitioning to osteoblasts, such as *Tcf7* (23g), *Bcl11b* (14g), and *Tbx2* (36g) (Fig. 8E [↗](#), Fig. 8-Supplementary Figure 1C [↗](#)). STRING network analysis showed that the genes with the strongest interaction with the osteo-core TFs are mostly related to Wnt signaling (Fig. 8F [↗](#)). This reveals the role of Wnt signaling in this transition from early fibrogenic activation of pSSPCs to osteogenic differentiation during bone repair. We calculated the osteo-core activity and observed that it is gradually increased and maintained in osteogenic cells, showing that the osteo-core is required for the transition and maturation of IIFC into osteoblasts (Fig. 8G-H [↗](#)). Overall, we identified distinct cores of regulons with distinct dynamics driving the transition of IIFCs into chondrocytes and osteoblasts.

IIFCs mediate paracrine interactions during bone repair

Paracrine cell interactions are crucial drivers of tissue regeneration and stem cell activation. To identify key cell interactions during bone repair, we performed cell interaction analyses using CellChat package³³ [↗](#). We observed that IIFCs are one of the predominant sources of outgoing signals during bone repair, and are also important receivers of signals, suggesting their central role in mediating cell interactions after fracture (Fig. 9A-B [↗](#)). Endothelial cells were mostly receiving signaling, while chondrocytes, osteoblasts and most immune cells exhibited reduced interactions with the other cell types in the fracture environment. IIFCs interact with all cell populations in the fracture environment, but the strongest interactions were with SSPCs and IIFCs (Fig. 9-Supplementary Figure 1A [↗](#)). CellChat analyses of the subset of SSPCs, IIFCs, osteoblasts and chondrocytes confirmed that IIFCs are a major source and receiver population of paracrine signals after fracture (Fig. 9-Supplementary Figure 1B [↗](#)). We then analyzed the main secreted factors from IIFCs. IIFCs secreted periostin (*Postn*), BMPs (*Bmp5*), pleiotropin/PTN (*Ptn*), TGFβs (*Tgfb2*, *Tgfb3*), PDGFs (*Pdgfc*, *Pdgfd*) and angiopoietin-likes/ANGPTLs (*Angplt2*, *Angplt4*) (Fig. 9C-D [↗](#), Fig. 9-Supplementary Figure 1C [↗](#)). We observed differences in the dynamics of these factors, as some of them peak at day 3 post-fracture such as TGFβ, while others peak at day 5 such as BMP, POSTN, PTN and ANGPLT (Fig. 9E [↗](#)). We assessed the dynamics of ligand and receptor expression and observed that ligand expression was increased during the IIFC phase and specific of the fracture response (Fig. 9F [↗](#)). Receptor expression was high in both SSPCs and IIFCs, and

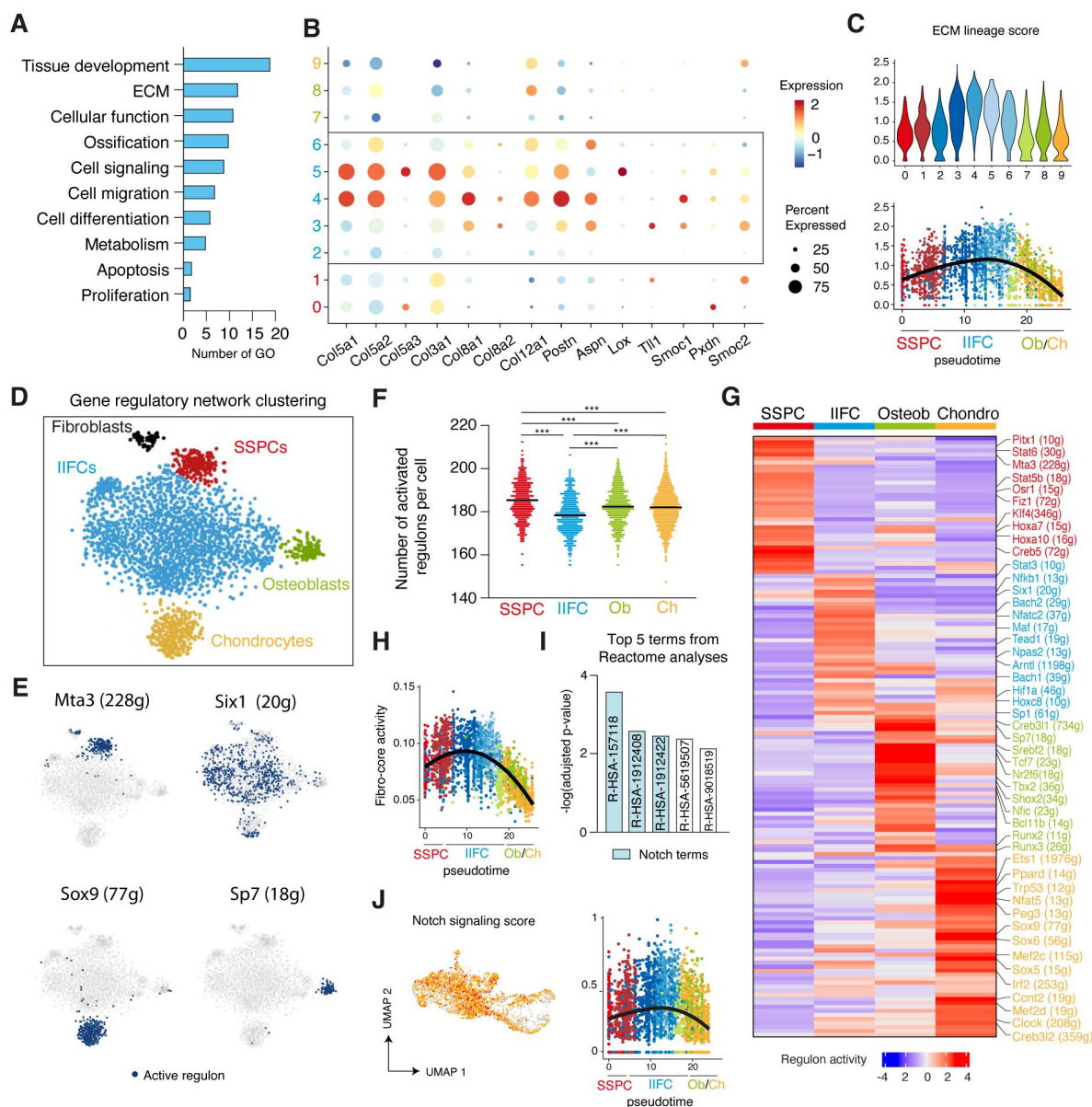


Figure 7

Characterization of injury-induced fibrogenic cells

A. Gene ontology analyses of upregulated genes in IIFCs (clusters 2 to 6 of UMAP clustering from Fig. 5C). **B.** Dot plot of ECM genes in UMAP clustering from Fig. 5C. **C.** Feature plot per cluster and scatter plot along pseudotime of the mean expression of ECM genes. **D.** Gene regulatory network (GRN)-based tSNE clustering of the subset of SSPCs, IIFCs, chondrocytes and osteoblasts. **E.** Activation of Mta3, Six1, Sox9 and Sp7 regulons in SSPCs, IIFCs, chondrocytes and osteoblasts. Blue dots mark cells with active regulon. **F.** Number of regulons activated per cell in the SSPC, IIFC, osteoblast (Ob) and chondrocyte (Ch) populations. Statistical differences were calculated using one-way ANOVA. ***: p-value < 0.001. **G.** Heatmap of activated regulons in SSPC, IIFC, osteoblast (osteob) and chondrocyte (chondro) populations. **H.** Scatter plot of the activity of the combined fibrogenic regulons along monocle pseudotime from Fig. 5C. **I.** Reactome pathway analyses of the fibrogenic regulons shows that the 3 most significant terms are related to Notch signaling (blue). **J.** Feature plot in Seurat clustering and scatter plot along monocle pseudotime of the Notch signaling score.

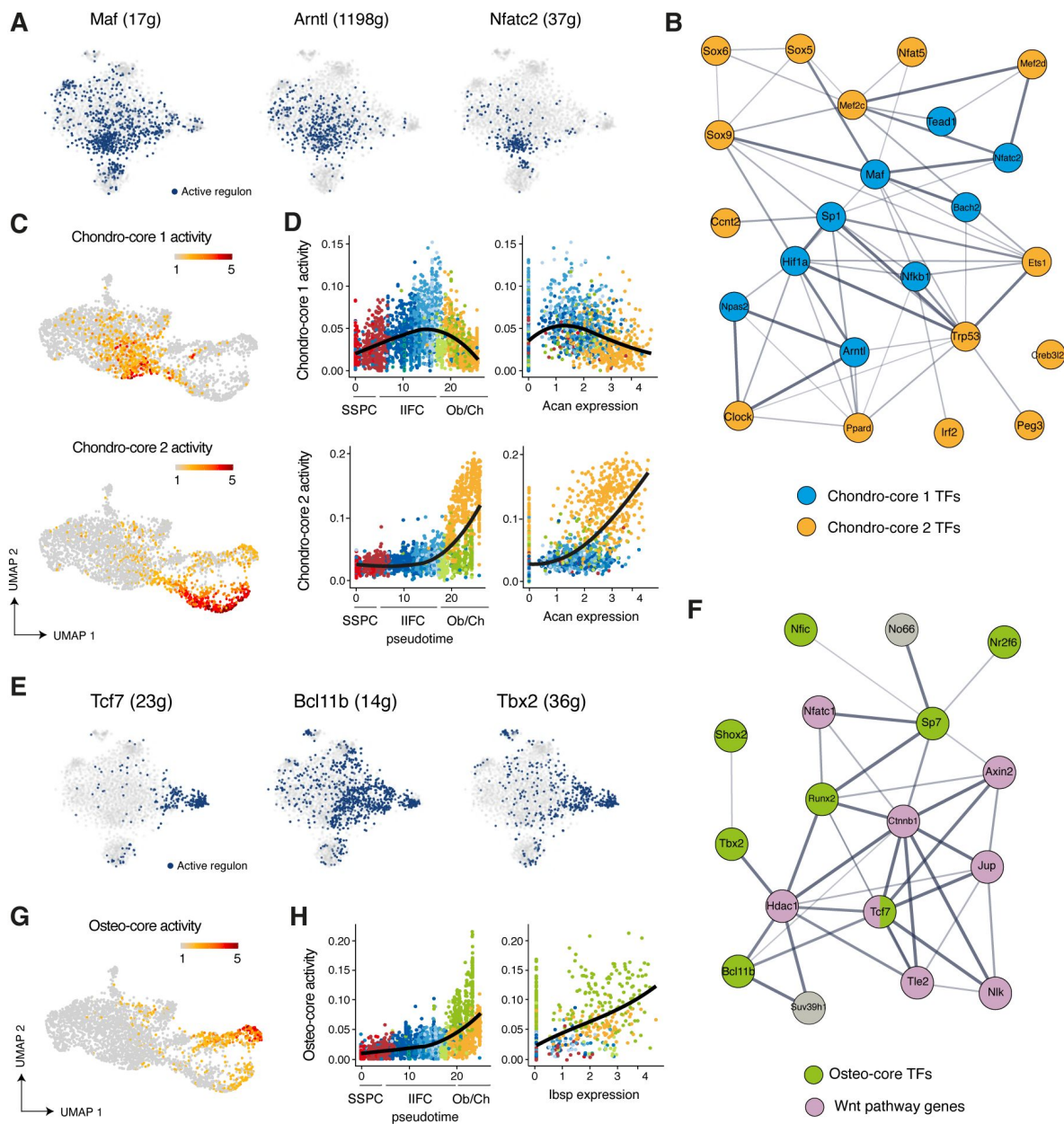


Figure 8

Gene regulatory network analyses identify gene cores driving fibrogenic to chondrogenic and osteogenic transitions.

A. Activation of Maf, Arntl, and Nfatc2 regulons in SSPCs, IIFCs, chondrocytes and osteoblasts. **B.** STRING interaction network of the chondro-core 1 and 2 transcription factors (blue and orange respectively). **C.** Feature plot of chondro-core 1 (top) and chondro-core 2 (bottom) activities in SSPCs, IIFCs, chondrocytes and osteoblasts in Seurat UMAP from [Fig. 5](#). **D.** Scatter plot of chondro-core 1 (top) and chondro-core 2 (bottom) activities along monocle pseudotime and *Acan* expressing. **E.** Activation of Tcf7, Bcl11b and Tbx2 regulons in SSPCs, IIFCs, chondrocytes and osteoblasts. **F.** STRING interaction network of the osteo-core transcription factors (green) and their related genes shows that most of osteo-core related genes are involved in Wnt pathway (purple). **G.** Feature plot of the osteo-core activity in SSPCs, IIFCs, chondrocytes and osteoblasts in Seurat UMAP from [Fig. 5](#). **H.** Scatter plot of osteo-core activity along monocle pseudotime and *Ibsp* expressing.

receptors were expressed from steady state, suggesting that SSPCs can receive signals from IIFCs. Analysis of SSPC incoming signals showed that IIFCs produce paracrine factors that can regulate SSPCs (**Fig. 9G**), indicating that they contribute to SSPC recruitment after fracture.

Discussion

The periosteum is the main driver of bone regeneration. Yet the periosteum composition and its response to bone fracture are poorly described. Here, we used single-nuclei transcriptomics to understand the heterogeneity of the periosteum at steady state and the changes occurring within the periosteum after bone fracture. We developed a protocol to extract nuclei from freshly dissected periosteum, allowing us to capture their intact transcriptomic profile without enzymatic digestion- induced stress^{13,34,35}. In addition, we performed snRNAseq without sorting specific populations, allowing the identification of all cell types located in the periosteum and the fracture environment from the early stages of repair. Our study provides the first complete fracture repair atlas, a key tool to understand bone regeneration.

First, we described the heterogeneity of the periosteum at steady state. While previous studies performed scRNAseq on sorted periosteal cell populations, our dataset uncovers the diversity of cell populations within the periosteum. We describe fibroblast populations, as well as tissue resident immune cells, adipocytes, and blood vessel/nerve-related cells. We identified one SSPC population of undifferentiated cells localized in the fibrous layer of the periosteum and expressing markers such as *SCA1*, *Dpp4* and *Pi16*, known to mark fibroblasts with stemness potential³⁶. No markers previously described such as *Ctsk*, or *Gli1* were fully specific of one cell cluster and of the undifferentiated SSPC population, suggesting that these markers may label heterogeneous cell populations^{2,5,6,19}.

After fracture, the composition of the periosteum changes drastically, with the appearance of IIFC and immune cells from day 3 post-fracture, and of chondrocytes and osteoblasts from day 5 post-fracture. Previous studies based on in vitro analyses and in vivo lineage tracing demonstrated that the periosteum displays the unique potential to form both bone and cartilage after fracture^{1,3,5}. Yet, it was still unknown if the periosteum contains a bipotent SSPC population or several SSPC populations with distinct potentials after injury. Here, we show that pSSPCs respond to injury and form bone and cartilage via a single trajectory emerging from *SCA1*⁺ SSPCs. Periosteal *SCA1*⁺ SSPCs become IIFCs, a state marked by a decreased expression of stemness markers and a strong expression of extracellular matrix genes, such as *Postn* and *Aspn* and activation of Notch- related TFs. Thus, bone injury does not induce an expansion of *SCA1*⁺ SSPCs, rather a transition toward IIFCs that progressively expand and represent the main cell population within the fracture callus until day 7 post-fracture. Following this fibrogenic step, IIFCs do not undergo cell death but undergo either osteogenesis or chondrogenesis. This newly identified transient IIFC state represents the crossroad of bone regeneration and SSPC differentiation. In tissues such as skeletal muscle, an early transient fibrogenic response is also required for regeneration supporting muscle stem cell activation and differentiation³⁷. During bone repair, this initial fibrogenic process is an integral part of the SSPC differentiation process, and a transitional step prior to osteogenesis and chondrogenesis.

GRN analyses identified TFs regulating SSPC response to fracture and involved in several cardinal signaling pathways including Notch and Wnt. Previous studies reported the role of Notch and Wnt in bone repair^{8,38-48}. Notch inactivation at early stages of repair leads to bone non-union while Notch inactivation in chondrocytes and osteoblasts does not significantly affect healing, correlating with our data showing that Notch is crucial during the IIFC phase, prior to osteochondral commitment³⁹. Our results show that Wnt activation occurs after Notch activation and is specific to IIFCs engaging into osteogenesis, confirming its crucial role as a regulator of osteogenesis^{8,45-48}. In addition, we identified a chondro-core of 9 regulators

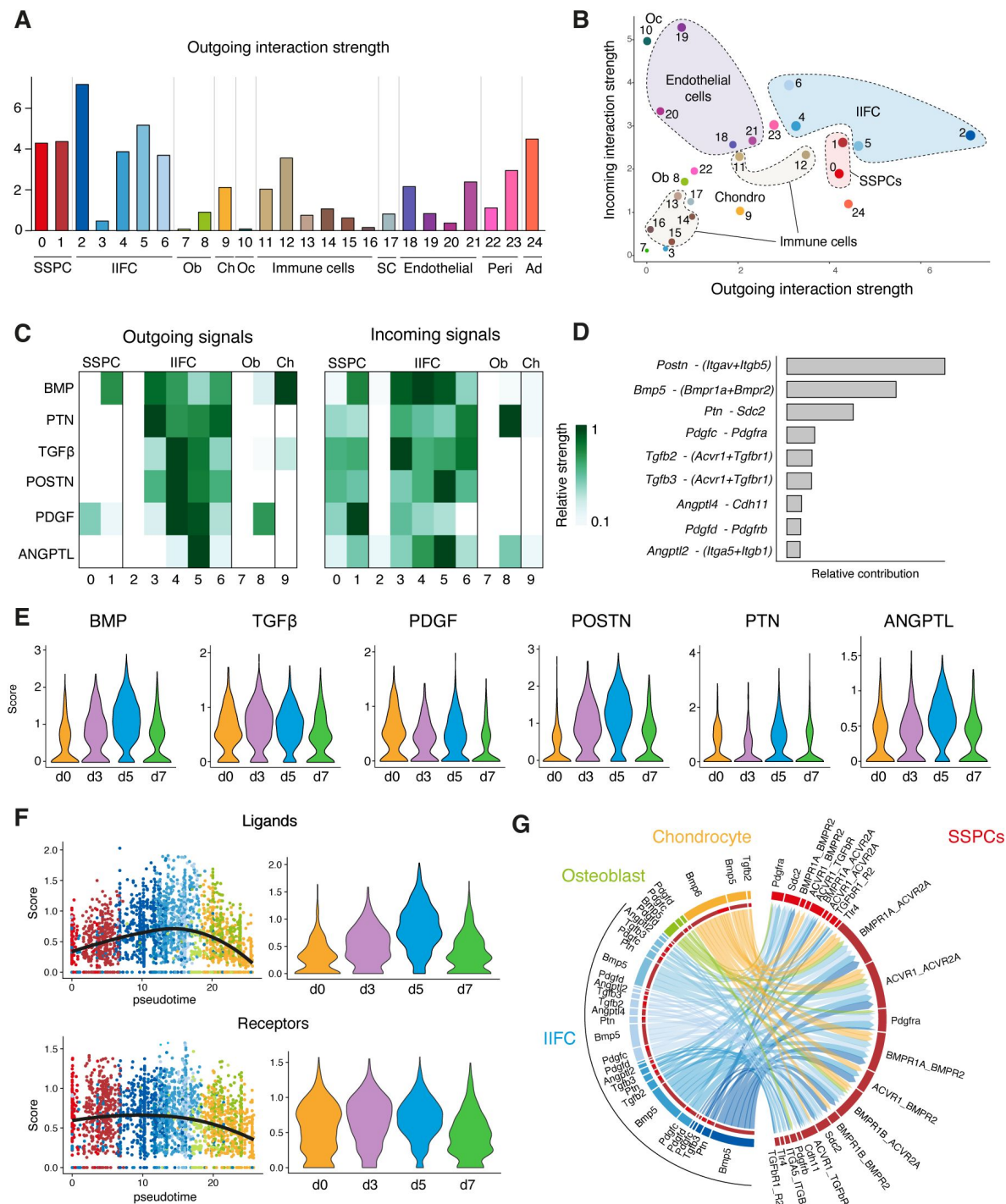


Figure 9

IIFCs are the main source of paracrine factors after fracture.

A. Outgoing interaction strengths of the different cell populations of the fracture environment determined using CellChat package. **B.** Comparison of incoming and outgoing interaction strengths across SSPC, IIFC, chondrogenic and osteogenic populations. **C.** Outgoing and incoming signaling from and to SSPCs, IIFCs, chondrocytes and osteoblasts. **D.** Cell-cell interactions identified between SSPCs, IIFCs, chondrocytes and osteoblasts. **E.** Violin plots of the score of BMP, TGFβ, PDGF, POSTN, PTN and ANGPTL signaling per timepoint. **F.** Scatter plot along pseudotime and violin plot per time point of the mean expression of the ligand and receptors involved in signaling from IIFCs. **G.** Circle plot of the interactions between SSPCs, IIFCs, chondrocytes and osteoblasts, showing that most signals received by SSPCs are coming for IIFCs. Ob: osteoblasts, Oc: osteoclasts, Ch: chondrocytes, SC: Schwann cells, Ad: Adipocytes.

transiently activated when IIFCs engage into chondrogenesis. Among these regulons, several are involved in the autonomous circadian clock, including *Npas2* and *Arntl* (Bmal1). Bmal1 was previously shown to be a regulator of cartilage differentiation in bone development and homeostasis⁴⁹. Bmal1 inactivation leads to chondrocyte apoptosis and disruption of signaling involved in cartilage formation, including TGF β , Ihh, HIF1 α and NFAT^{50–53}. This suggests a role of the circadian clock genes as key regulators of chondrogenesis during bone repair. IIFCs are also revealed as crucial paracrine regulators of the fracture environment. IIFCs secrete factors including BMPs, PDGFs, TGF β s, and POSTN, known to be required for successful bone healing^{1,3,18,54}. These signals potentially regulate the interactions between IIFCs and all other cell types in the fracture environment, but primarily SSPCs and IIFCs themselves. Thus, IIFCs appear to contribute to the recruitment of SSPCs and promote their own maturation and differentiation.

Overall, our study provides a complete dataset of the early steps of bone regeneration. The newly identified SSPC activation pattern involves a coordinated temporal dynamic of cell phenotypes and signaling pathways after injury. IIFCs emerge as a transient cell population that plays essential roles in the initial steps of bone repair. Deeper understanding of IIFC regulation will be crucial as they represent the ideal target to enhance bone healing and potentially treat bone repair defects.

Materials and methods

Mice

C57BL/6ScNj were obtained from Janvier Labs (France). *Prx1*^{Cre55} and *Rosa26-mtdTomato-mEGFP* (*R26*^{mTmG})⁵⁶ were obtained from Jackson Laboratory (Bar Harbor, ME). All SSPCs, including pSSPCs, are marked by GFP in *Prx1*^{Cre}; *R26*^{mTmG} mice. Mice were bred in animal facilities at IMRB, Creteil and kept in separated ventilated cages, in pathogen-controlled environment and ad libitum access to water and food. All procedures performed were approved by the Paris Est Creteil University Ethical Committee. Twelve-week-old males and females were mixed in experimental groups.

Tibial fracture

Open non-stabilized tibial fractures were induced as previously described⁵⁷. Mice were anesthetized with an intraperitoneal injection of Ketamine (50 mg/mL) and Medetomidine (1 mg/kg) and received a subcutaneous injection of Buprenorphine (0.1 mg/kg) for analgesia. Mice were kept on a 37°C heating pad during anesthesia. The right hindlimb was shaved and sanitized. The skin was incised to expose the tibia and osteotomy was performed in the mid-diaphysis by cutting the bone. The wound was sutured, the mice were revived with an intraperitoneal injection of atipamezole (1 mg/kg) and received two additional analgesic injections in the 24 hours following surgery. Mice were sacrificed at 3, 5, 7 or 14 days post-fracture.

Nuclei extraction

Nuclei extraction protocol was adapted from^{58,59}. We generated 4 datasets for this study: uninjured periosteum, periosteum and hematoma at days 3, 5 and 7 post-tibial fracture. The uninjured and day 3 post-fracture datasets were generated in duplicates. For uninjured periosteum, tibias from 4 mice were dissected free of muscle and surrounding tissues. The epiphyses were cut and the bone marrow flushed. The periosteum was scraped from the cortex using dissecting Chisel (10095-12, Fine Science Tools). For days 3, 5 and 7 post fracture, injured tibias from 4 to 9 mice were collected and the surrounded tissues were removed. The activated periosteum was scraped and collected with the hematoma. Collected tissues were minced and placed 5 min in ice-cold Nuclei Buffer (NUC101, Merck) before mechanical nuclei extraction using a glass douncer. Extraction was performed by 20 strokes of pestle A followed by 5-10 of pestle B.

Nuclei suspension was filtered, centrifuged and resuspended in RNase-free PBS (AM9624, ThermoFischer Scientific) with 2% Bovine Serum Albumin (A2153, Merck) and 0.2 U/ μ L RNase inhibitor (3335399001, Roche). A second step of centrifugation was performed to reduce contamination by cytoplasmic RNA. Sytox™ AADvanced™ (S10349, ThermoFischer Scientific) was added (1/200) to label nuclei and Sytox- AAD+ nuclei were sorted using Sony SH800.

Single nuclei RNA sequencing

The snRNA-seq libraries were generated using Chromium Single Cell Next GEM 3' Library & Gel Bead Kit v3.1 (10x Genomics) according to the manufacturer's protocol. Briefly, 10 000 to 20 000 nuclei were loaded in the 10x Chromium Controller to generate single-nuclei gel-beads in emulsion. After reverse transcription, gel-beads in emulsion were disrupted. Barcoded complementary DNA was isolated and amplified by PCR. Following fragmentation, end repair and A-tailing, sample indexes were added during index PCR. The purified libraries were sequenced on a Novaseq (Illumina) with 28 cycles of read 1, 8 cycles of i7 index and 91 cycles of read 2. Sequencing data were processed using the Cell Ranger Count pipeline and reads were mapped on the mm10 reference mouse genome with intronic and exonic sequences.

Filtering and clustering using Seurat

Single-nuclei RNAseq analyses were performed using Seurat v4.1.0 ⁶⁰[60](#),⁶¹[61](#) and Rstudio v1.4.1717. Aligned snRNAseq datasets were filtered to retain only nuclei expressing between 200 et 5000 genes and expressing less than 2% of mitochondrial genes and 1.5% of ribosomal genes. Contamination from myogenic cells were removed from the analyses. After filtering, we obtained 1378 nuclei from uninjured periosteum, 1634 from day 3 post-fracture, 2089 from d5 post-fracture and 1112 from day 7 post-fracture. The replicates of the uninjured dataset were integrated using Seurat. The integrated dataset was regressed on cell cycle, mitochondrial and ribosomal content and clustering was performed using the first 15 principal components and a resolution of 0.5. SSPC/fibroblast and osteogenic cells were isolated and reclustered using the first 10 principal components and a resolution of 0.2. Uninjured, d3, d5 and d7 datasets were integrated using Seurat. The integrated dataset was regressed on cell cycle, mitochondrial and ribosomal content. Clustering was performed using the first 20 principal components and a resolution of 1.3. SSPC, IIFC, chondrogenic and osteogenic clusters from the integration were isolated to perform subset analysis. The subset was reclustered using the first 15 principal components and a resolution of 0.6.

Pseudotime analysis using monocle3

Monocle3 v1.0.0 was used for pseudotime analysis ⁶²[62](#). Single-cell trajectories were determined using monocle3 default parameters. The starting point of the pseudotime trajectory was determined as the cells from the uninjured dataset with the highest expression of stem/progenitor marker genes (*Ly6a*, *Cd34*, *Dpp4*, *Pi16*). Pseudotime values were added in the Seurat object as metadata and used with the Seurat package.

Differentiation state analysis using CytoTRACE

To assess the level of differentiation of the cell clusters, we performed analyses using CytoTRACE with the default parameters. CytoTRACE scoring was plotted in violin plot and on the Seurat UMAP clustering.

Gene Ontology and Reactome and analyses

Reactome and GO analyses were performed using EnrichR ⁶³[63](#). All significant GO terms from upregulated genes in clusters 2 to 6 of the subset of SSPCs, IIFCs, osteoblasts and chondrocytes were manually categorized. The 5 more significant terms of the Reactome analysis from the fibrogenic TFs of **Fig. 6** [6](#) are presented in **Table S3** [S3](#).

Single cell regulatory network inference using SCENIC

Single cell regulatory network inference and clustering (SCENIC) ²² was used to infer transcription factor (TF) networks active in SSPCs, IIFCs, osteoblasts and chondrocytes. Analysis was performed using recommended parameters using the packages SCENIC v1.3.1, AUCell v1.16.0, and RcisTarget v1.14 and the motif databases RcisTarget and GRNboost. SCENIC package was used to perform regulon based tSNE clustering and identified population specific regulons.

Cell-cell interaction using CellChat

Cell communication analysis was performed using the R package CellChat ³³ with default parameters on the complete fracture combined dataset and on the subset of SSPCs, IIFCs, osteoblasts and chondrocytes.

STRING network analyses

To assess protein-protein interaction network, we used the STRING v11.5 database ⁶⁴. To assess interaction in the chondro-core, we performed the analysis on the chondro-core and chondro-specific TFs identified in our analysis. For osteo-core analyses, we performed the analysis with osteo-core genes and the most significant interactions.

Histology and immunofluorescence

Mouse samples were processed as previously described ⁵⁷. Tibias were collected and fixed in 4% PFA (sc-281692, CliniSciences) for 4 hours at 4°C. Then, samples were decalcified in 19% EDTA for 10 days (EU00084, Euromedex), cryoprotected in 30% sucrose (200-301-B, Euromedex) for 24h and embedded in OCT. Samples were sectioned in 10µm thick sections. Cryosections were defrosted and rehydrated in PBS. For Safranin-O staining, sections were stained with Weigert's solution for 5 min, rinsed in running tap water for 3 min and stained with 0.02% Fast Green for 30 seconds (F7252, Merck), followed by 1% acetic acid for 30 seconds and Safranin'O solution for 45 min (S2255, Merck). For immunofluorescence, sections were incubated 1 hour at room temperature in 5% serum, 0.25% Triton PBS before incubation overnight at 4°C with the following antibodies: Rat monoclonal to mouse SCA1 (740450, BD Biosciences), Rabbit monoclonal to mouse SOX9 (ab185230, Abcam), Rabbit polyclonal to mouse Osterix/Sp7 (ab22552, Abcam), Goat polyclonal to mouse Periostin (AF2955, R&D Systems), Goat polyclonal to mouse PECAM1 (AF3628, Biotechne), Rat monoclonal to mouse CD45 (552848, BD Bioscience). Secondary antibody incubation was performed at room temperature for 1 hour. Slides were mounted with Fluoromount- G mounting medium with DAPI (00-4959-52, Life Technologies).

RNAscope in situ hybridization

The expression of *Pi16* and *Postn* was visualized using the RNAscope® Multiplex Fluorescent Assay V2 (Biotechne). Tissues were processed as described above. Ten µm thick sections were cut and processed according to the manufacturer's protocol: 15 min of post-fixation in 4% PFA, ethanol dehydration, 10 min of H2O2 treatment and incubation in ACD custom reagent for 30 minutes at 40°C. After hybridization and revelation, the sections were mounted under a glass coverslip with Prolong Gold Antifade (P10144, ThermoFischer).

Tissue dissociation and cell sorting

Periosteal SSPC isolation

To isolate periosteal cells, uninjured tibias from *Prx1^{Cre}; R26^{mTmG}* mice were collected, and all surrounding soft tissues were carefully removed. Epiphyses were embedded in low melting agarose and tibias were placed for 30 at 37°C in digestion medium composed of PBS with 3mg/ml

of Collagenase B (C6885, Merck), 4mg/ml of Dispase (D4693, Merck) and 100U/mL of DNase I (WOLS02007, Serlabo, France). After digestion, tibias were removed and the suspension was filtered, centrifuged and resuspended.

Injury-induced fibrogenic cell isolation

The fracture hematoma and the activated periosteum were collected from *Prx1^{Cre}; R26^{mTmG}* mice 3 days post-fracture. Tissues were minced and digested at 37°C for 2 hours in DMEM (21063029, Life Technologies) with 1% Trypsin (15090046, Life Technologies) and 1% collagenase D (11088866001, Roche). Cells in suspension were removed every 20 min and the digestion medium was replaced. After 2 hours, the cell suspension was filtered, centrifuged and resuspended.

Cell sorting and transplantation

Digested cells were incubated 30 minutes with anti-SCA1-BV650 (BD Biosciences, 740450) or anti-CD146-BV605 (BD Biosciences, 740636). Cells were sorted using Influx Cell Sorter. Prior sorting, Sytox Blue (S34857, Thermo Fisher Scientific) was added to stain dead cells. We sorted living single GFP⁺ SCA1⁺ (pSSPCs excluding GFP⁺ endothelial cells and SCA1⁺ pericytes) and living single GFP⁺ SCA1⁺ cells from the uninjured periosteum of *Prx1^{Cre}; Rosa^{mTmG}* mice. For IIFCs, living single GFP⁺ CD146⁺ cells were sorted, to eliminate CD146⁺ pericytes. Cell transplantation was performed as described in [57](#), 30000 to 45000 sorted GFP⁺ cells were embedded in Tisseel Prima fibrin gel, composed of fibrinogen and thrombin (3400894252443, Baxter S.A.S, USA), according to manufacturer's instructions. Briefly, the cells were resuspended in 15 µL of fibrin (diluted at 1/4), before adding 15 µL of thrombin (diluted at 1/4) and mixing. The pellet was then placed on ice for at least 15 min for polymerization. The cell pellet was transplanted at the fracture site of wild-type mice at the time of fracture.

In vitro colony-forming Unit (CFU) assay

Prx1-GFP⁺ SCA1⁺ and Prx1-GFP⁺ SCA1⁺ periosteal cells were isolated from *Prx1^{Cre}; Rosa^{mTmG}* mice as described above. Sorted cells were plated at a cell density of 2000 cells per 6 well-plates in growth media consisting of MEMa supplemented with 20% FBS, 1% penicillin-streptomycin (Life Technology, Carlsbad, California), and 0.01% bFGF (R&D, Minneapolis, MN). The medium was changed every 2-3 days. After 11 days, plates were washed twice with PBS, fixed with methanol, stained with 0,5% crystal violet diluted in 20% methanol and colonies were manually counted.

Statistical analyses

Statistical difference between the number of activated regulons per cell was determined using one way-ANOVA followed by a post-hoc test. A p-value < 0.001 is reported as ***.

Resource availability

Single nuclei RNAseq datasets are deposited in the Gene Expression Omnibus (GSE234451). The integrated dataset was deposited on USCS Cell Browser for easy use: (<https://fracture-repair-atlas.cells.ucsc.edu>). This paper does not report original code. Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Céline Colnot (celine.colnot@inserm.fr).

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Additional information

Conflict of interests

Authors declare no competing interests.

Author contributions

Conceptualization: S.P. and C.C. Methodology: S.P. and C.C. Formal analysis: S.P. and C-A.W. Investigation: S.P., M.E., V.B., C.G., M.L., F.C. and C.M. Resources: F.C. and M.M. Writing – Original Draft: S.P. and C.C. Visualization: S.P. Supervision: C.C. Project administration: C.C. Funding Acquisition: C.C.

Supplemental information

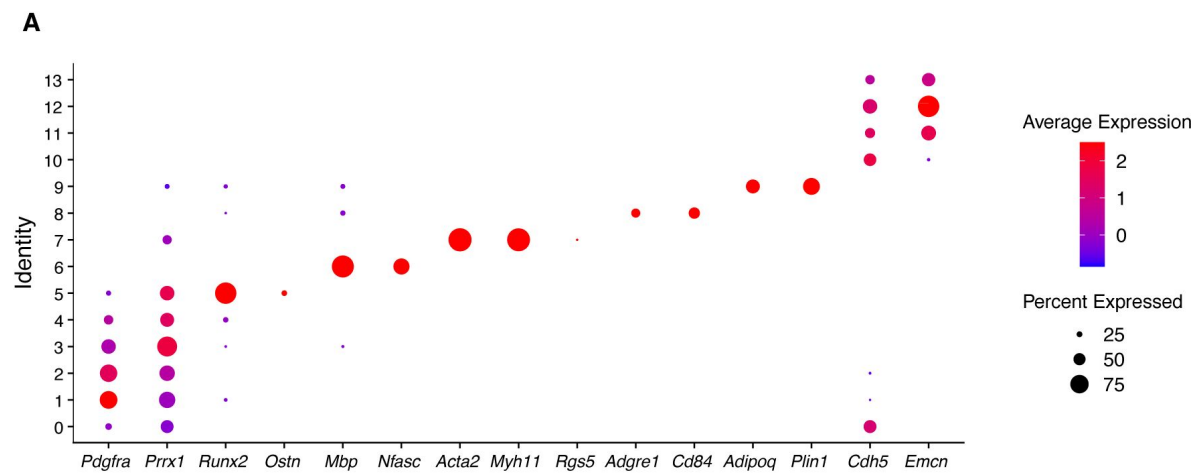


Figure 1 – Supplementary Figure 1A

Dot plot of marker genes of the populations from uninjured periosteum

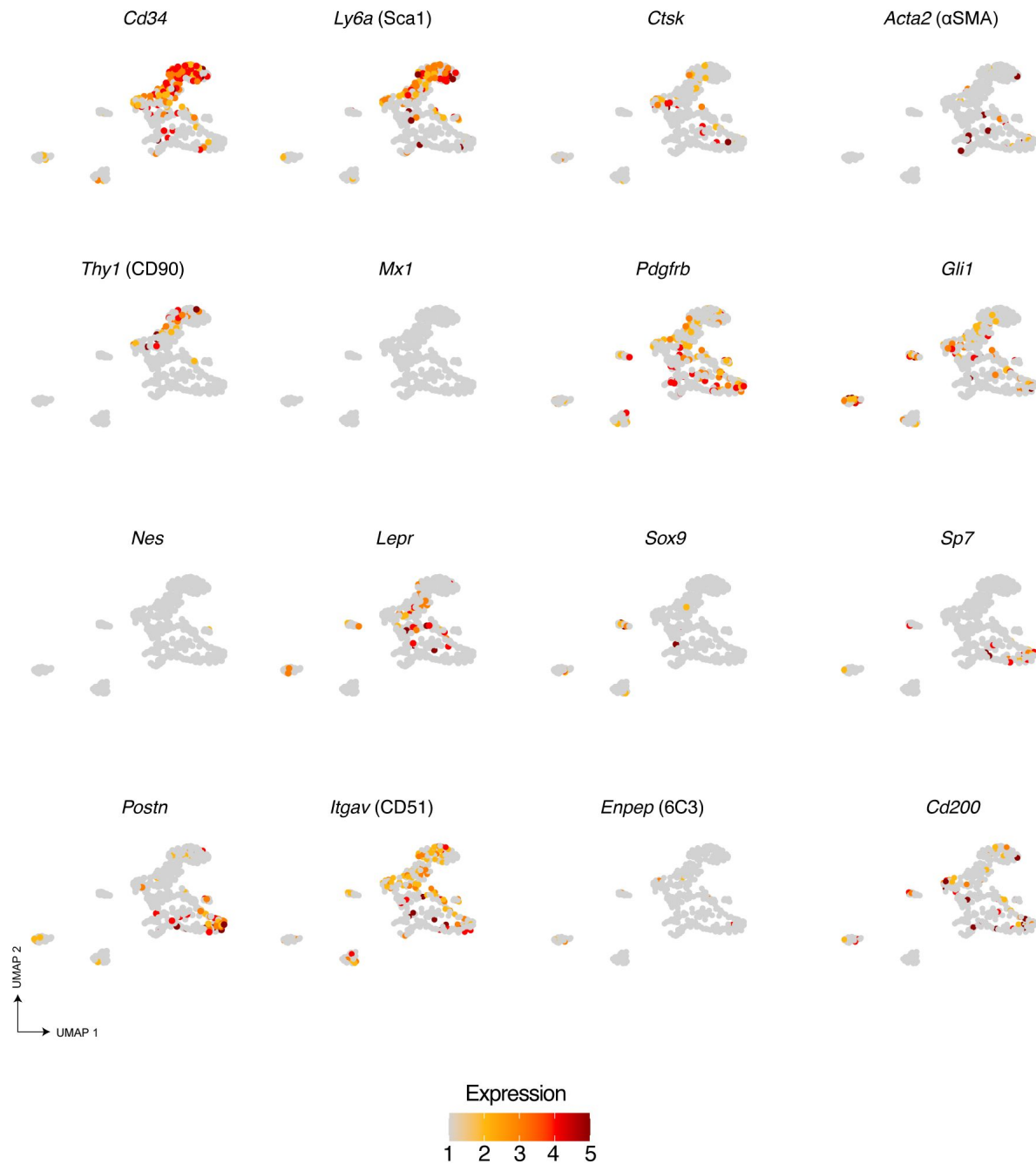


Figure 2 – Supplementary Figure 1

Expression of known SSPC markers in the periosteum at steady state.

Feature plots of known markers of SSPCs in the SSPC/fibroblast subset in periosteum at steady state.

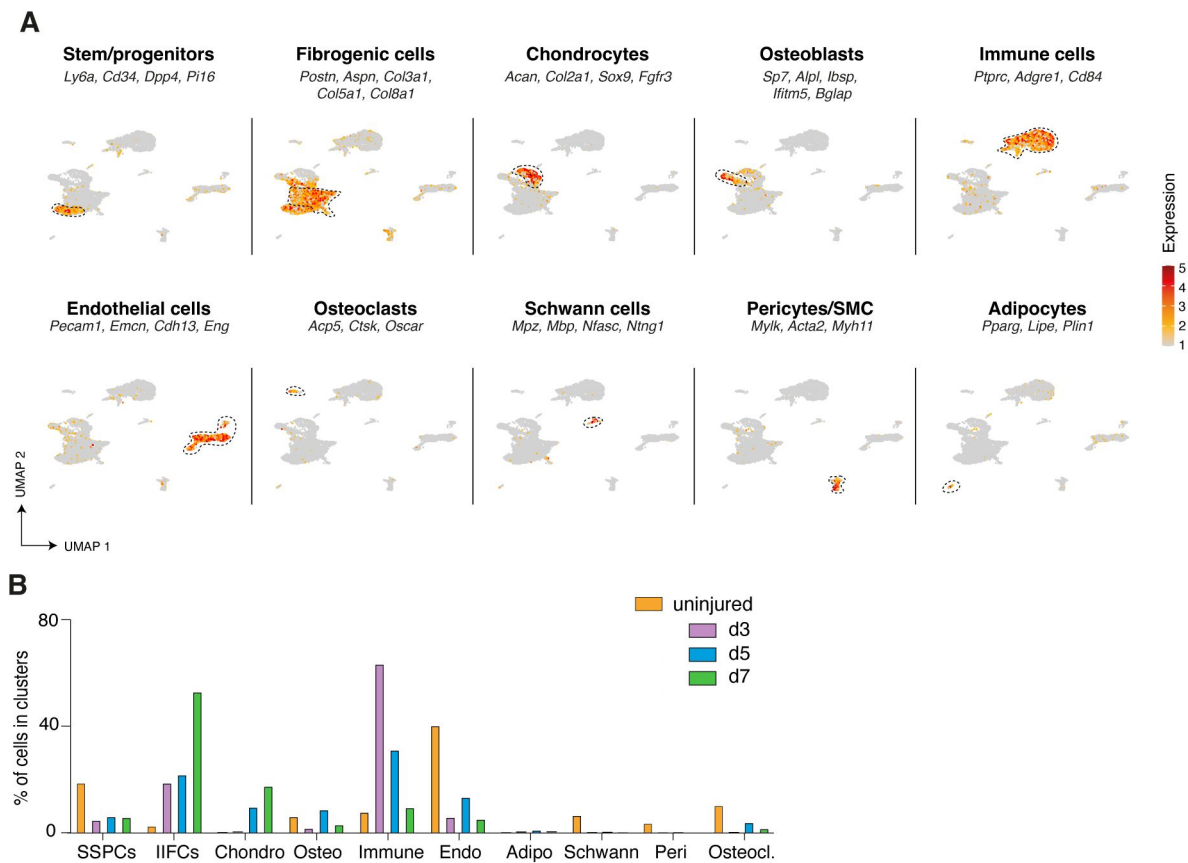


Figure 3 – Supplementary Figure 1

Heterogeneity and dynamics of the cell populations in the fracture environment.

A. Feature plots of the lineage score of the different cell populations in the combined fracture datasets. **B.** Percentage of cells in each cell population per time point.

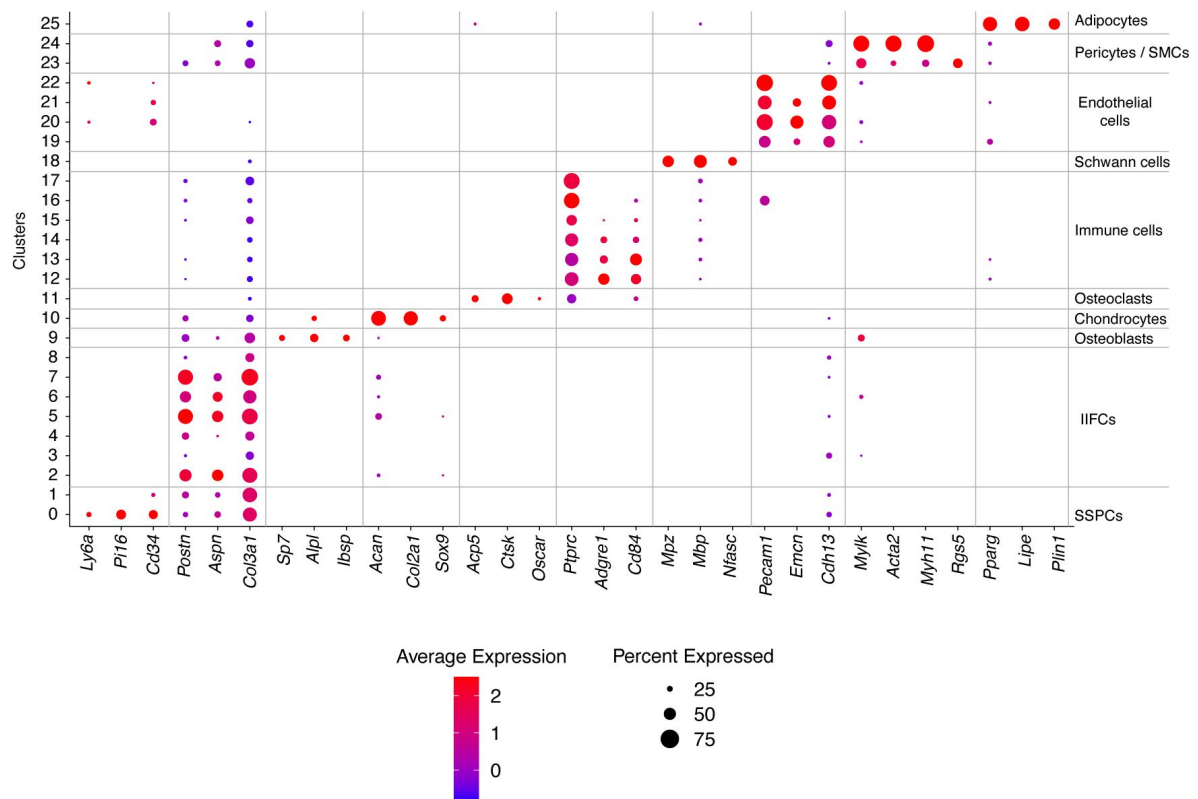


Figure 3 - Supplementary Figure 2

Dot plot of marker genes of the populations from the combined fracture dataset

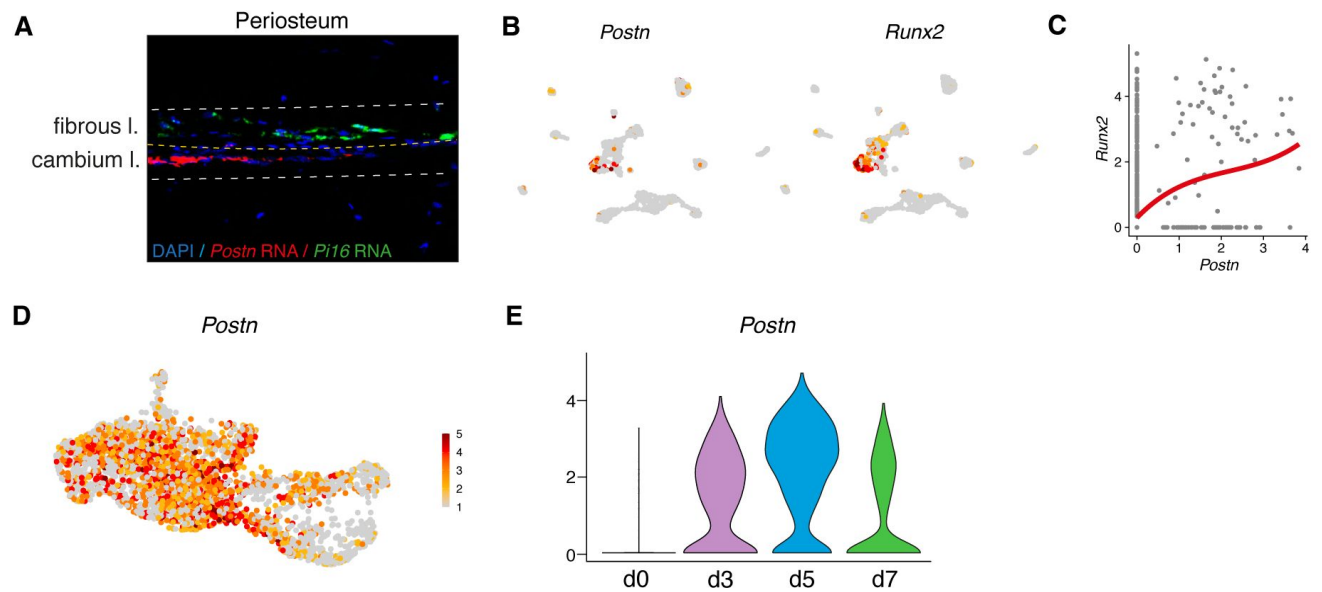


Figure 4 – Supplementary Figure 1

Periosteal SSPCs do not express *Postn*

A. RNAscope experiment showing the presence of *Postn* expressing cells in the inner cambium layer (cl) of the periosteum and *Pi16*-expressing cells in the fibrous layer (fl). **B.** Feature plots of *Postn* and *Runx2* expression in the uninjured periosteum. **C.** Scatter plot of *Runx2* and *Postn* expression in the uninjured periosteum dataset showing that *Postn* is mostly expressed by cells expressing *Runx2*. **D.** Feature plot of *Postn* expression in the subset of SSPCs, IIFCs, osteogenic and chondrogenic cells from [Fig. 5B](#). **E.** Violin plot of *Postn* expression per time point.

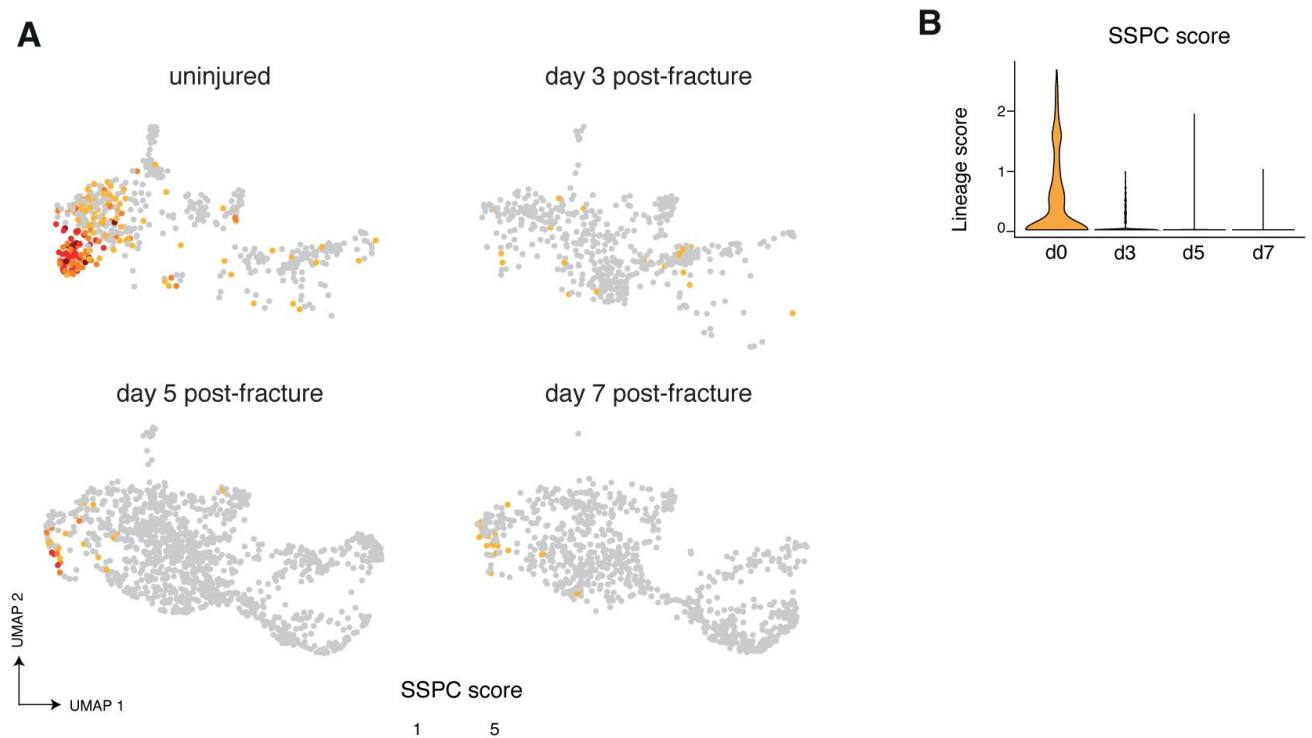


Figure 4 – Supplementary Figure 2

Absence of SSPCs in the injured periosteum.

A. Feature plot of SSPC lineage score in the subset of SSPCs, IIFCs, osteoblasts and chondrocytes separated by time point from **Fig. 5B**. **B.** Violin plot of SSPC lineage score by time point.

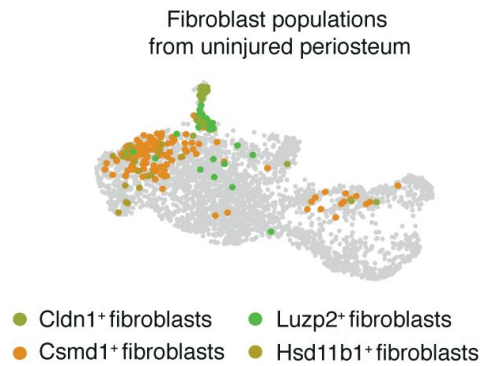


Figure 5 – Supplementary Figure 1

UMAP projection highlighting the distribution of periosteal fibroblasts in the combined fracture dataset.

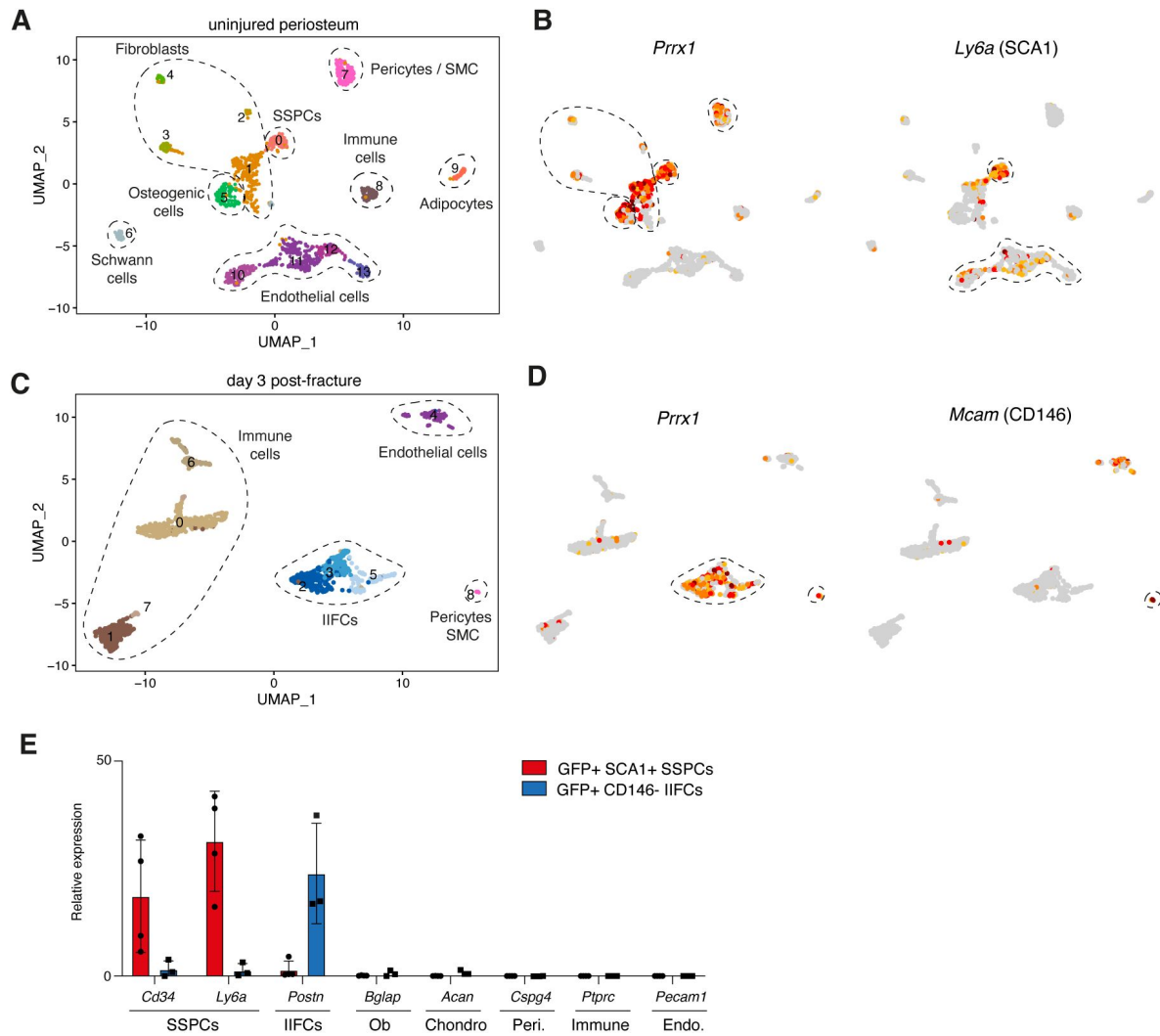


Figure 6 – Supplementary Figure 1

Validation of SSPC and IIFC sorting strategies.

A. UMAP projection of the clustering of the uninjured periosteum. **B.** Feature plots of *Prrx1* and *Ly6a* expression in the uninjured dataset. **C.** UMAP projection of the clustering of the day 3 post-fracture periosteum and hematoma. **D.** Feature plots of *Prrx1* and *Mcam* (CD146) expression in the day 3 post-fracture periosteum and hematoma dataset. **E.** Relative expression of cell population markers by *Prx1*⁺ SCA1⁺ SSPCs from uninjured periosteum and *Prx1*⁺ CD146⁻ IIFCs from day 3 post-fracture periosteum and hematoma.

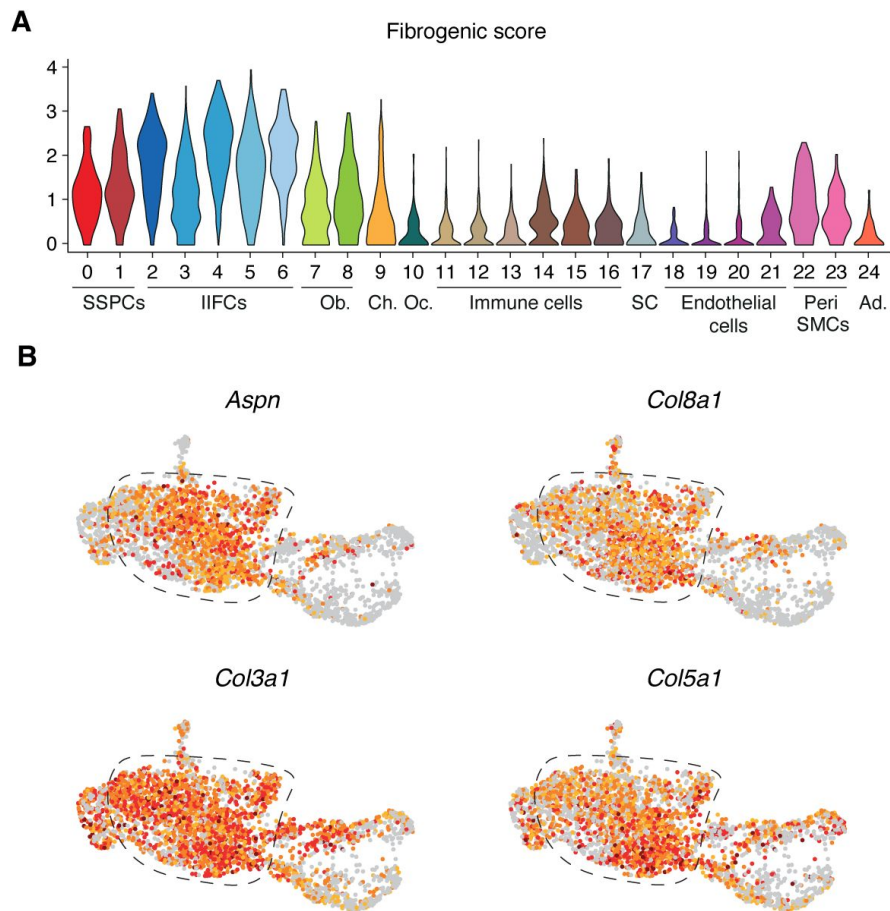


Figure 7 – Supplementary Figure 1

IIFCs are expressing ECM-related genes.

A. Violin plot of the extracellular matrix genes score in the integrated dataset. **B.** Feature plots of *Aspn*, *Col3a1*, *Col5a1* and *Col8a1* in the subset of SSPCs, IIFCs, osteoblasts and chondrocytes.

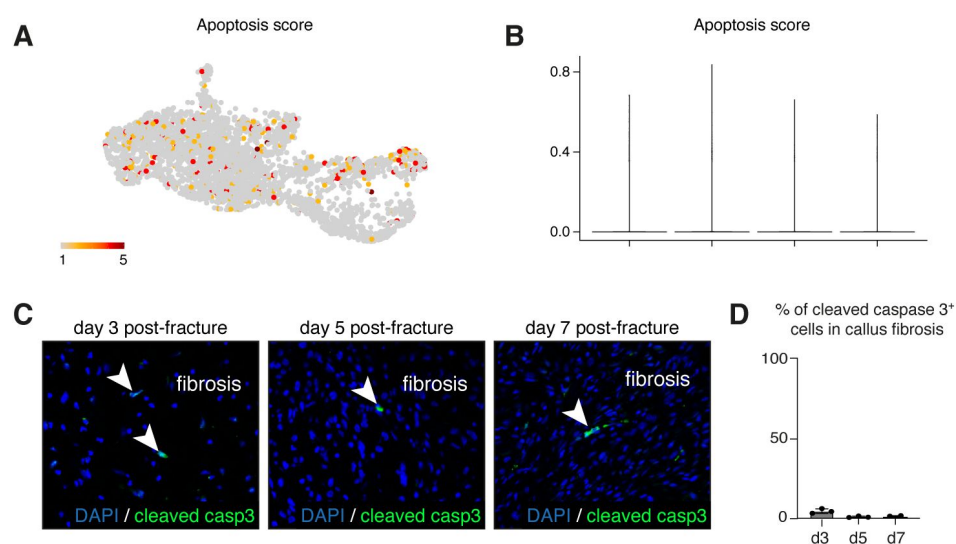


Figure 7 – Supplementary Figure 2

IIFCs do not undergo apoptosis.

A. Feature plot of the apoptosis score in the subset of SSPCs, IIFCs, osteoblast and chondrocytes. **B.** Violin plot of the apoptosis score separated by time points. **C.** Immunofluorescence of cleaved caspase 3 in callus fibrosis at day 3, 5 and 7 post-fracture. **D.** Percentage of cleaved caspase 3 positive cells in the fibrosis callus fibrosis at day 3, 5 and 7 post-fracture.

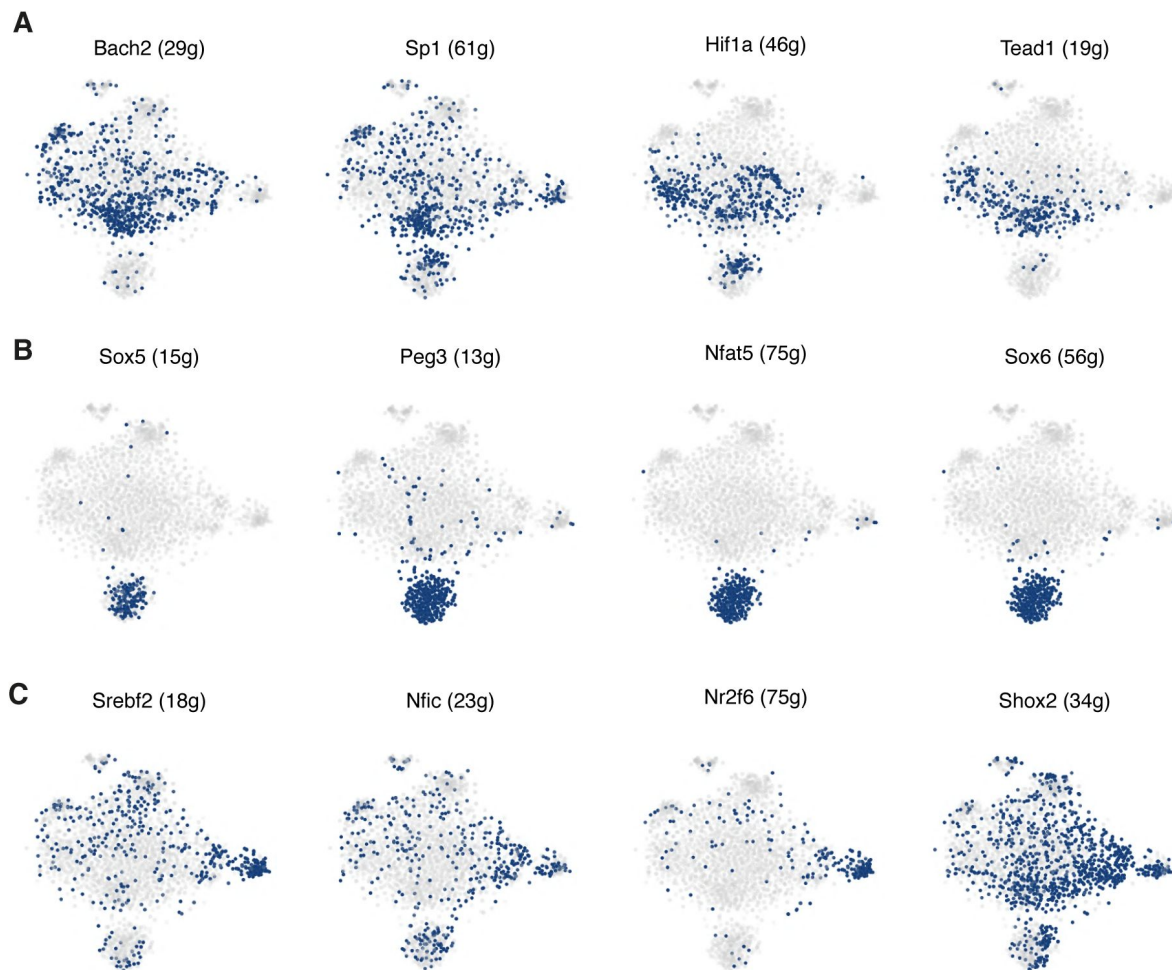


Figure 8 – Supplementary Figure 1

Regulon activity in the subset of SSPCs, IIFCs, osteoblasts and chondrocytes

A. Activity of Mta3, Six1, Sox9 and Sp7 regulons in the UMAP Seurat clustering of SSPCs, IIFCs, chondrocytes and osteoblasts. **B.** Activity of chondro-core 1 regulons in SCENIC tSNE clustering of SSPCs, IIFCs, chondrocytes and osteoblasts. **C.** Activity of chondro-core 2 regulons in SCENIC tSNE clustering of SSPCs, IIFCs, chondrocytes and osteoblasts. **D.** Activity of osteo-core regulons in SCENIC tSNE clustering of SSPCs, IIFCs, chondrocytes and osteoblasts.

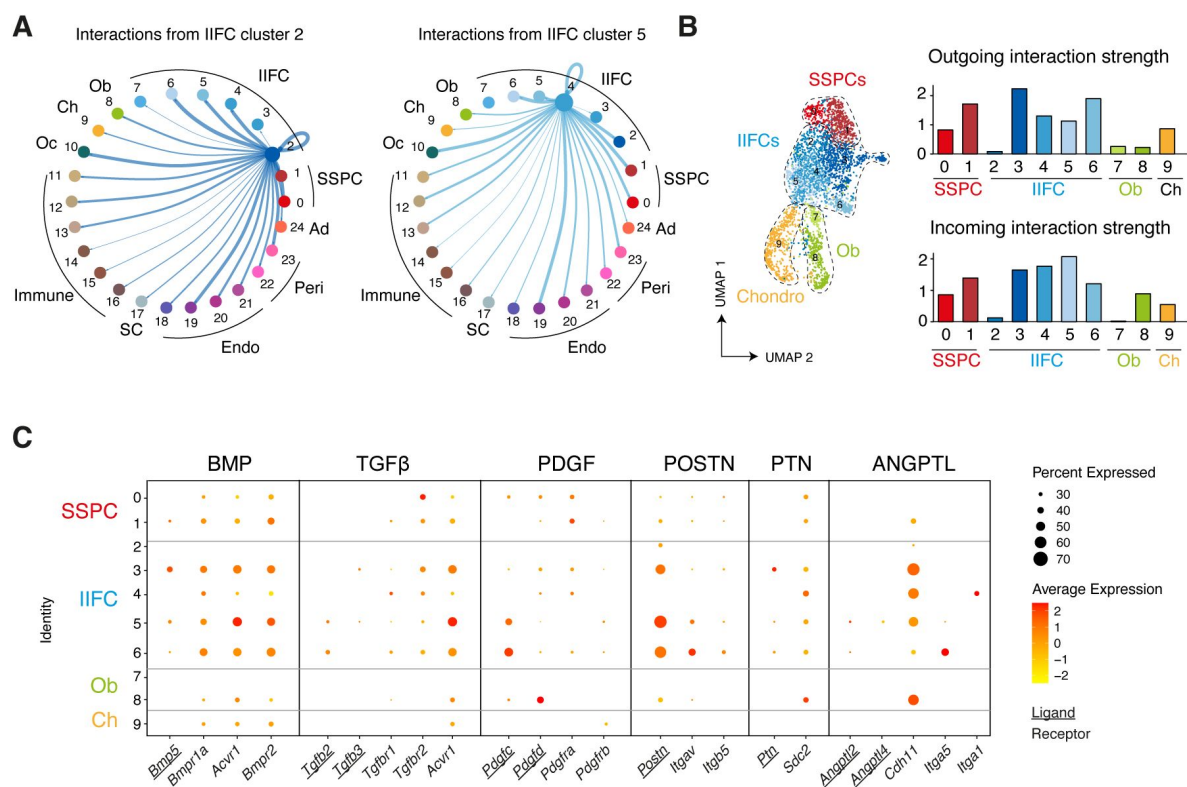


Figure 9 – Supplementary Figure 1

Paracrine interactions from IIFCs.

A. Circle plots showing the interaction strengths between IIFCs in clusters 2 and 5 with the other cell populations. **B.** (left) Feature plot of the subset of SSPCs, IIFCs, chondrocytes and osteoblasts from Fig 5. (right) Outgoing and incoming interaction strengths of the subset of SSPCs, IIFCs, chondrocytes and osteoblasts. **C.** Dot plots of the expression of the ligands (underlined) and receptors of BMP, TGF β , PDGF, POSTN, PTN and ANGPTL family involved in cell-cell interactions from IIFCs after fracture.

Score	Genes
Stem/progenitor score	<i>Ly6a, Cd34, Dpp4, Pi16</i>
Fibrogenic score	<i>Postn, Aspn, Col3a1, Col5a1, Col8a1</i>
Chondrogenic score	<i>Acan, Col2a1, Sox9, Fgfr3</i>
Osteogenic score	<i>Sp7, Alpl, Ibsp, Ifitm5, Bglap</i>
Notch score	<i>Notch1, Notch3, Notch2, Notch4, Jag1, Dtx3, Fbxw7, Psen1, Epn2, Sel1l, Maml1, Spen, Anxa4</i>
ECM score	<i>Col5a1, Col5a2, Col5a3, Col3a1, Col8a1, Col8a2, Col12a1, Eln, Postn, Aspn, Lox, Tll1, Smoc1, Pxdn, Smoc2</i>
BMP score	<i>Bmp5, Bmpr1a, Acvr1, Bmpr2</i>
TGFb score	<i>Tgfb2, Tgfb3, Tgfb1, Tgfb2, Acvr1</i>
PDGF score	<i>Pdgfc, Pdgfd, Pdgfra, Pdgfrb</i>
POSTN score	<i>Postn, Itgav, Itgb5</i>
PTN score	<i>Ptn, Sdc2</i>
ANGPLT score	<i>Angptl2, Angptl4, Cdh11, Itga1, Itga5</i>
Total ligand score	<i>Bmp5, Tgfb2, Tgfb3, Postn, Ptn, Angptl4, Angptl2, Pdgfc, Pdgfd</i>
Total receptor score	<i>Bmpr1a, Acvr1, Bmpr2, Tgfb1, Tgfb2, Itgav, Itgb5, Sdc2, Ncl, Pdgfra, Pdgfrb</i>

Table S1

Lists of genes used for lineage score analyses of murine snRNAseq.

Group of regulons	Regulons
Fibro-core regulons	<i>Arnt, Bach1, Bcl3, Cux1, E2f3, Elk4, Etv6, Fosl2, Foxn3, Foxp1, Jun, Junb, Mdb4, Meis1, Myc, Nfib, Pbx1, Pbx3, Rocor1, Runx1, Six1</i>
Chondro-core 1 regulons	<i>Npas2, Hif1a, Sp1, Arntl, Bach2, Nfkb1, Maf, Tead1, Nfatc2</i>
Chondro-core 2 regulons	<i>Sox6, Sox5, Sox9, Mef2c, Nfat5, Mef2d, Ccnt2, Clock, Ppard, Irf2, Trp53, Peg3, Creb3l2, Ets1</i>
Osteo-core regulons	<i>Nr2f6, Nfic, Shox2, Tbx2, Sp7, Bcl11b, Tcf7, Runx2</i>

Table S2

Lists of the regulons composing the cores.

Reactome number	Term	Adjusted p-value
R-HSA-157118	Signaling By NOTCH	2,52E-04
R-HSA-1912408	Pre-NOTCH Transcription And Translation	0,00250164
R-HSA-1912422	Pre-NOTCH Expression And Processing	0,00331873
R-HSA-5619507	Activation Of HOX Genes During Differentiation	0,00394002
R-HSA-9018519	Estrogen-dependent Gene Expression	0,00697148

Table S3

Top 5 terms from Reactome analysis on fibro-core regulons.

References

1. Duchamp de Lageneste O., et al. (2018) **Periosteum contains skeletal stem cells with high bone regenerative potential controlled by Periostin** *Nat Commun* **9**
2. Jeffery E. C., Mann T. L. A., Pool J. A., Zhao Z., Morrison S. J (2022) **Bone marrow and periosteal skeletal stem/progenitor cells make distinct contributions to bone maintenance and repair** *Cell Stem Cell* **29**:1547–1561
3. Julien A., et al. (2022) **Skeletal stem/progenitor cells in periosteum and skeletal muscle share a common molecular response to bone injury** *J of Bone & Mineral Res* <https://doi.org/10.1002/jbmr.4616>
4. Julien A., et al. (2021) **Direct contribution of skeletal muscle mesenchymal progenitors to bone repair** *Nature Communications* **12**
5. Debnath S., et al. (2018) **Discovery of a periosteal stem cell mediating intramembranous bone formation** *Nature* **562**:133–139
6. Matthews B. G., et al. (2021) **Heterogeneity of murine periosteum progenitors involved in fracture healing** *eLife* **10**
7. Ortinau L. C., et al. (2019) **Identification of Functionally Distinct Mx1+αSMA+ Periosteal Skeletal Stem Cells** *Cell Stem Cell* **25**:784–796
8. Matsushita Y., et al. (2020) **A Wnt-mediated transformation of the bone marrow stromal cell identity orchestrates skeletal regeneration** *Nat Commun* **11**
9. De Micheli A. J., et al. (2020) **Single-Cell Analysis of the Muscle Stem Cell Hierarchy Identifies Heterotypic Communication Signals Involved in Skeletal Muscle Regeneration** *Cell Rep* **30**:3583–3595
10. Guerrero-Juarez C. F., et al. (2019) **Single-cell analysis reveals fibroblast heterogeneity and myeloid- derived adipocyte progenitors in murine skin wounds** *Nat Commun* **10**
11. Storer M. A., et al. (2020) **Acquisition of a Unique Mesenchymal Precursor-like Blastema State Underlies Successful Adult Mammalian Digit Tip Regeneration** *Developmental Cell* **52**:509–524
12. Sivaraj K. K., et al. (2022) **Mesenchymal stromal cell-derived septoclasts resorb cartilage during developmental ossification and fracture healing** *Nat Commun* **13**
13. Machado L., et al. (2021) **Tissue damage induces a conserved stress response that initiates quiescent muscle stem cell activation** *Cell Stem Cell* **28**:1125–1135
14. Selewa A., et al. (2020) **Systematic Comparison of High-throughput Single-Cell and Single-Nucleus Transcriptomes during Cardiomyocyte Differentiation** *Sci Rep* **10**
15. Ding J., et al. (2020) **Systematic comparison of single-cell and single-nucleus RNA-sequencing methods** *Nat Biotechnol* **38**:737–746

16. Wen F., Tang X., Xu L., Qu H (2022) **Comparison of single-nucleus and single-cell transcriptomes in hepatocellular carcinoma tissue** *Mol Med Rep* **26**
17. Tournaire G., et al. (2020) **Nestin-GFP transgene labels skeletal progenitors in the periosteum** *Bone* **133**
18. Gao B., et al. (2019) **Macrophage-lineage TRAP+ cells recruit periosteum-derived cells for periosteal osteogenesis and regeneration** *Journal of Clinical Investigation* **129**:2578–2594
19. Chan C. K. F., et al. (2015) **Identification and Specification of the Mouse Skeletal Stem Cell** *Cell* **160**:285–298
20. He X., et al. (2017) **Sox9 positive periosteal cells in fracture repair of the adult mammalian long bone** *Bone* **103**:12–19
21. Böhm A.-M., et al. (2019) **Activation of Skeletal Stem and Progenitor Cells for Bone Regeneration Is Driven by PDGFR β Signaling** *Developmental Cell* **51**:236–254
22. Aibar S., et al. (2017) **SCENIC: single-cell regulatory network inference and clustering** *Nat Methods* **14**:1083–1086
23. Leclerc K., et al. (2023) **Hox genes are crucial regulators of periosteal stem cell identity** *Development* **150**
24. Nemec S., et al. (2017) **Pitx1 directly modulates the core limb development program to implement hindlimb identity** *Development* **144**:3325–3335
25. Takahashi K., Yamanaka S (2006) **Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors** *Cell* **126**:663–676
26. Chen H., et al. (2011) **Activation of STAT6 by STING is critical for antiviral innate immunity** *Cell* **147**:436–446
27. Kollmann S., et al. (2021) **A STAT5B-CD9 axis determines self-renewal in hematopoietic and leukemic stem cells** *Blood* **138**:2347–2359
28. Tsuruyama T., et al. (2010) **Dual retrovirus integration tagging: identification of new signaling molecules Fiz1 and Hipk2 that are involved in the IL-7 signaling pathway in B lymphoblastic lymphomas** *J Leukoc Biol* **88**:107–116
29. Le Grand F., et al. (2012) **Six1 regulates stem cell repair potential and self-renewal during skeletal muscle regeneration** *J Cell Biol* **198**:815–832
30. Linares A. J., et al. (2015) **The splicing regulator PTBP1 controls the activity of the transcription factor Pbx1 during neuronal differentiation** *eLife* **4**
31. Rottkamp C. A., Lobur K. J., Wladyka C. L., Lucky A. K., O’Gorman S (2008) **Pbx3 is required for normal locomotion and dorsal horn development** *Dev Biol* **314**:23–39
32. Wang S., et al. (2023) **Meis1 Controls the Differentiation of Eye Progenitor Cells and the Formation of Posterior Poles during Planarian Regeneration** *Int J Mol Sci* **24**
33. Jin S., et al. (2021) **Inference and analysis of cell-cell communication using CellChat** *Nat Commun* **12**

34. Ding J., et al. (2020) **Systematic comparison of single-cell and single-nucleus RNA-sequencing methods** *Nat Biotechnol* **38**:737–746
35. van den Brink S. C., et al. (2017) **Single-cell sequencing reveals dissociation-induced gene expression in tissue subpopulations** *Nat Methods* **14**:935–936
36. Buechler M. B., et al. (2021) **Cross-tissue organization of the fibroblast lineage** *Nature* **593**:575–579
37. Wosczyzna M. N., et al. (2019) **Mesenchymal Stromal Cells Are Required for Regeneration and Homeostatic Maintenance of Skeletal Muscle** *Cell Rep* **27**:2029–2035
38. Lee S., et al. (2021) **Notch-Wnt signal crosstalk regulates proliferation and differentiation of osteoprogenitor cells during intramembranous bone healing** *npj Regen Med* **29**
39. Wang C., et al. (2016) **NOTCH signaling in skeletal progenitors is critical for fracture repair** *Journal of Clinical Investigation* **126**:1471–1481
40. Dishowitz M. I., Terkhorn S. P., Bostic S. A., Hankenson K. D (2012) **Notch signaling components are upregulated during both endochondral and intramembranous bone regeneration** *Journal Orthopaedic Research* **30**:296–303
41. Kraus J. M., et al. (2022) **Notch signaling enhances bone regeneration in the zebrafish mandible** *Development* **149**
42. Matthews B. G., et al. (2014) **Analysis of α SMA-Labeled Progenitor Cell Commitment Identifies Notch Signaling as an Important Pathway in Fracture Healing** *J Bone Miner Res* **29**:1283–1294
43. Novak S., et al. (2020) **Modulation of Notch1 signaling regulates bone fracture healing** *J Orthop Res* **38**:2350–2361
44. Cao J., et al. (2017) **Notch signaling pathway promotes osteogenic differentiation of mesenchymal stem cells by enhancing BMP9/Smad signaling** *International Journal of Molecular Medicine* **40**:378–388
45. Minear S., Leucht P., Miller S., Helms J. A (2010) **rBMP represses Wnt signaling and influences skeletal progenitor cell fate specification during bone repair** *J Bone Miner Res* **25**:1196–1207
46. Minear S., et al. (2010) **Wnt Proteins Promote Bone Regeneration** *Sci. Transl. Med* **2**
47. Kang S., et al. (2007) **Wnt Signaling Stimulates Osteoblastogenesis of Mesenchymal Precursors by Suppressing CCAAT/Enhancer-binding Protein α and Peroxisome Proliferator-activated Receptor γ** *Journal of Biological Chemistry* **282**:14515–14524
48. Komatsu D. E., et al. (2010) **Modulation of Wnt signaling influences fracture repair** *Journal Orthopaedic Research* **28**:928–936
49. Chen G., et al. (2020) **The biological function of BMAL1 in skeleton development and disorders** *Life Sciences* **253**
50. Dudek M., et al. (2023) **The clock transcription factor BMAL1 is a key regulator of extracellular matrix homeostasis and cell fate in the intervertebral disc** *bioRxiv* <https://doi.org/10.1101/2023.02.12.528214>

51. Dudek M., et al. (2015) **The chondrocyte clock gene Bmal1 controls cartilage homeostasis and integrity** *Journal of Clinical Investigation* **126**:365–376
52. Ma Z., et al. (2019) **Deletion of clock gene Bmal1 impaired the chondrocyte function due to disruption of the HIF1 α -VEGF signaling pathway** *Cell Cycle* **18**:1473–1489
53. Takarada T., et al. (2012) **Clock Genes Influence Gene Expression in Growth Plate and Endochondral Ossification in Mice** *Journal of Biological Chemistry* **287**:36081–36095
54. Xia C., et al. (2020) **TGF- β /Smad2 signalling regulates enchondral bone formation of Gli1 + periosteal cells during fracture healing** *Cell Prolif* **53**
55. Logan M., et al. (2002) **Expression of Cre recombinase in the developing mouse limb bud driven by aPrxl enhancer** *genesis* :77–80
56. Muzumdar M. D., Tasic B., Miyamichi K., Li L., Luo L (2007) **A global double-fluorescent Cre reporter mouse** *Genesis* **45**:593–605
57. Perrin S., Julien A., Duchamp de Lageneste O., Abou-Khalil R., Colnot C. (2021) **Mouse Periosteal Cell Culture, in vitro Differentiation, and in vivo Transplantation in Tibial Fractures** *Bio-protocol* **11**
58. Santos M. D., et al. (2021) **Extraction and sequencing of single nuclei from murine skeletal muscles** *STAR Protocols* **2**
59. Martelotto L. G. (2019) **‘Frankenstein’ protocol for nuclei isolation from fresh and frozen tissue for snRNAseq v2** *protocols.io* <https://doi.org/10.17504/protocols.io.3fkgjkw>
60. Stuart T., et al. (2019) **Comprehensive Integration of Single-Cell Data** *Cell* **177**:1888–1902
61. Butler A., Hoffman P., Smibert P., Papalexi E., Satija R (2018) **Integrating single-cell transcriptomic data across different conditions, technologies, and species** *Nat Biotechnol* **36**:411–420
62. Cao J., et al. (2019) **The single-cell transcriptional landscape of mammalian organogenesis** *Nature* **566**:496–502
63. Kuleshov M. V., et al. (2016) **Enrichr: a comprehensive gene set enrichment analysis web server 2016 update** *Nucleic Acids Res* **44**:W90–97
64. Szklarczyk D., et al. (2021) **The STRING database in 2021: customizable protein-protein networks, and functional characterization of user-uploaded gene/measurement sets** *Nucleic Acids Res* **49**:D605–D612

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Reviewer #1 (Public review):

This study delineates an important set of uninjured and injured periosteal snRNAseq data that provides an overview of periosteal cell responses to fracture healing. The authors also took additional steps to validate some of the findings using immunohistochemistry and transplantation assays. This study will provide a valuable publicly accessible dataset to reexamine the expression of the reported periosteal stem and progenitor cell markers.

Strengths:

- (1) This is the first single-nuclei atlas of periosteal cells that are obtained without enzymatic cell dissociation or targeted cell purification by FACS. This integrated snRNAseq dataset will provide additional opportunities for the community to revisit the expression of many periosteal cell markers that have been reported to date.
- (2) The authors delved further into the dataset using cutting-edge algorithms, including CytoTrace, SCENIC, Monocle, STRING and CellChat, to define potential roles of identified cell populations in the context of fracture healing. These additional computation analyses generate many new hypotheses regarding periosteal cell reactions.
- (3) The authors also sought to validate some of the computational findings using immunohistochemistry and transplantation assays to support the conclusion.

Weaknesses:

- (1) The current snRNAseq datasets contain only a small number of nuclei (1,189 nuclei at day 0, 6,213 nuclei day 0-7 combined). It is possible that these datasets are underpowered to discern subtle biological changes in skeletal stem/progenitor cell populations during fracture healing.
- (2) POSTN is expressed in the cambium layer of the periosteum without fracture. The current data do not exclude the possibility that these pre-existing POSTN+ cells are the main responder of fracture healing.

<https://doi.org/10.7554/eLife.92519.2.sa2>

Reviewer #2 (Public review):

Summary:

The authors described cell type mapping was conducted for both WT and fracture types. Through this, unique cell populations specific to fracture conditions were identified. To determine these, the most undifferentiated cells were initially targeted using stemness-related markers and CytoTrace scoring. This led to the identification of SSPC differentiating into fibroblasts. It was observed that the fibroblast cell type significantly increased under fracture conditions, followed by subsequent increases in chondrocytes and osteoblasts.

Strengths:

This study presented the injury-induced fibrogenic cell (IIFC) as a characteristic cell type appearing in the bone regeneration process and proposed that the IIFC is a progenitor undergoing osteochondrogenic differentiation.

Comments on revised version:

The authors have thoroughly addressed the reviewer's comments and have conducted additional experiments.

<https://doi.org/10.7554/eLife.92519.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

This study delineates an important set of uninjured and injured periosteal snRNAseq data that provides an overview of periosteal cell responses to fracture healing. The authors also took additional steps to validate some of the findings using immunohistochemistry and transplantation assays. This study will provide a valuable publicly accessible dataset to reexamine the expression of the reported periosteal stem and progenitor cell markers.

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(2) The authors delved further into the dataset using cutting-edge algorithms, including CytoTrace, SCENIC, Monocle, STRING, and CellChat, to define the potential roles of identified cell populations in the context of fracture healing. These additional computation analyses generate many new hypotheses regarding periosteal cell reactions.

(3) The authors also sought to validate some of the computational findings using immunohistochemistry and transplantation assays to support the conclusion.

Weaknesses:

(1) The current snRNAseq datasets contain only a small number of nuclei (1,189 nuclei at day 0, 6,213 nuclei on day 0-7 combined). It is unclear if the number is sufficient to discern subtle biological processes such as stem cell differentiation.

We analyzed a total of 6,213 nuclei from uninjured periosteum and fracture calluses at 3 stages of bone healing. We were able to describe 11 distinct cell populations, revealing the diversity of cell populations in uninjured periosteum and post-injury, including rare cell types in the fracture environment such as Schwann cells, adipocytes and pericytes. The number of nuclei was sufficient to perform extensive analysis using a combination of cutting-edge algorithms. We agree that more nuclei would allow more in-depth analyses of cell fate transitions and rare populations, such as pericytes and Schwann cells. However, we concentrated here on SSPC/fibrogenic cells that are well represented in our dataset. Our study robustness is also reinforced by the analysis of 4 successive time points to define the SSPC/fibrogenic cell trajectories. Our validations using immunohistochemistry and transplantation assays also confirmed that our dataset is sufficient to define cell trajectories. There is no clear consensus on the number of cells needed to perform sc/snRNAseq analyses, as it depends on the cell types analyzed and the fold changes in gene expression. Previously reported single cell datasets containing a lower number of cells reached major conclusions

including SSPC identification, cell differentiation trajectories and differential gene expression (658 cells in (Debnath et al. 2018), 300 in (Ambrosi et al. 2021), around 175 in (Remark et al. 2023).)

(2) The authors' designation of Sca1+CD34+ cells as SSPCs is not sufficiently supported by experimental evidence. It will be essential to demonstrate stem/progenitor properties of Sca1+CD34+ cells using independent biological approaches such as CFU-F assays. In addition, the putative lineage trajectory of SSPCs toward IIFCs, osteoblasts, and chondrocytes remains highly speculative without concrete supporting data.

We performed additional analyses to further support that Sca1+ SSPCs display stem/progenitor properties. We performed CFU assays with Prx1-GFP+ SCA1+ and Prx1-GFP+ SCA1- periosteal cells (Figure 2F-G). We showed that Prx1-GFP+ SCA1+ display significant increased CFU potential compared to Prx1-GFP+ SCA1- cells. In addition, we isolated and transplanted Prx1-GFP+ Sca1+ and Prx1-GFP+ Sca1- periosteal cells at the fracture site of wild-type mice (Figure 2H). Only Sca1+ cells contributed to the callus formation, reinforcing that Sca1+ cells are the SSPC population mediating bone repair.

The differentiation trajectory of SSPCs presented in our study is supported by a combination of bioinformatic analyses and in vivo validation:

- snRNAseq allowed us to identify the different populations in the uninjured periosteum. In silico, in vitro and in vivo analyses all point to Sca1+ cells as the SSPC population (Fig 2EG).

- At day 3 post-fracture, we did not detect Sca1+ cells in the callus (Fig 4 – Supplementary figure 2). Instead, we observed the appearance of a new population, IIFCs. This population clustered along SSPCs and pseudotime analyses indicate that SSPCs can differentiate into IIFCs (Fig 5B). We confirmed the ability of Sca1+ pSSPCs to form IIFCs, by grafting them in the fracture callus and assessing their fibrogenic fate at day 5 post-fracture (Fig 6B).

- In silico, we observed that IIFCs clustered along osteogenic and chondrogenic cells. The pseudotime trajectory suggests that IIFCs can differentiate into both lineages (Fig 5B-C). This is coherent with the progressive expression of osteochondrogenic genes observed in IIFCs (Fig 5C, Fig 8A, C, E). In vivo, we observed the progressive expression of Runx2 and Sox9 by IIFCs undergoing differentiation (Fig 6A). We now show that IIFCs are not undergoing apoptosis, indicating that these cells further differentiate (Fig 7 – Supplementary figure 2). To functionally assess the osteochondrogenic potential of IIFCs, we used transplantation assay and showed that Prx1-GFP+ IIFCs isolated from day 3 post-fracture form cartilage and bone when transplanted at the fracture site of wild-type mice (Fig 6C).

We would like to insist on the robustness of the bioinformatic analyses performed in our study. First, we used datasets from different time points post-fracture to capture the true temporal progression of cell populations in the fracture callus. We used a large combination of tools shown to be reliable in many studies (Julien et al. 2021; Matsushita et al. 2020; Debnath et al. 2018; Baccin et al. 2020; Junyue Cao et al. 2019; Zhong et al. 2020), and all tools converge in the same trajectory. To further show the relevance of pseudotime in our model, we illustrated the distribution of the cell populations by time point (Fig. 5D). We can observe a parallel between the time points and the pseudotime, reinforcing that the pseudotime trajectory reflects the timing of SSPC differentiation. Overall, the combined in silico, in vitro and in vivo analyses support that Sca1+ Pi16+ cells are the periosteal SSPC population, specifically represented in the uninjured dataset. In response to bone fracture, these SSPCs give rise to IIFCs that are specifically represented in the intermediate stages (days 3 and 5) prior to osteochondrogenic differentiation.

(3) The designation of POSTN+ clusters as injury-induced fibrogenic cells (IIFCs) is not fully supported by the presented data. The authors' snRNAseq datasets (Figure 1d)

demonstrate that there are many POSTN+ cells prior to injury, indicating that POSTN+ cells are not specifically induced in response to injury. It has been widely recognized that POSTN is expressed in the periosteum without fracture. This raises a possibility that the main responder of fracture healing is POSTN+ cells, not SSPCs as they postulate. The authors cannot exclude the possibility that Sca1+CD34+ cells are mere bystanders and do not participate in fracture healing.

IIFCs are a population of cells that express high levels of ECM related genes, including *Postn*, *Aspn* and collagens. We did not claim that *Postn* expression is specific to IIFCs. While *Postn* is detected in the uninjured periosteum, snRNAseq analyses and RNAscope experiments showed that the expression of *Postn* is limited to a small number of cells in the cambium layer of the periosteum (Fig 4B, Figure 4 – Supplementary figure 1B). These *Postn*-expressing cells in the uninjured periosteum are not SSPCs, as they do not co-express/co-localize with Pi16+ and Sca1+ cells detected in the fibrous layer (Fig4, Figure 4– Supplementary figure 1A, Figure 6-Supplementary figure 1). These *Postn*-expressing cells are undergoing osteogenic differentiation as shown by the correlation between *Runx2* and *Postn* expression (Fig. 4 – Supplementary Figure 1C). After fracture, we observed a strong increase in ECM-related gene expression and specifically in the IIFC population. We now show the strong increase of *Postn* expression after injury (Fig. 4 – Supplementary Figure 1D-E, Figure 6-Supplementary figure 1E).

As mentioned in our response above, we now show that SCA1+ cells form cartilage and bone after fracture, while SCA1- cells (including the POSTN+ population) from the uninjured periosteum did not contribute. These data reveal that Sca1+ CD34+ cells are the main SSPC population mediating bone healing and that POSTN+ IIFCs are a transient stage of SSPC differentiation. We added the following text to the result section: “Pi16-expressing SSPCs are located within the fibrous layer, while we observed few POSTN+ cells in the cambium layer (Fig. 4 – Supplementary Fig. 1A). *Postn* expression is weak in uninjured periosteum and is limited to differentiating cells. *Postn* expression is strongly increased in response to fracture, specifically in IIFCs (Fig. 4 – Supplementary Fig. 1B-E). “

(4) Detailed spatial organization of Sca1+CD34+ cells and POSTN+ cells in the uninjured periosteum with respect to the cambium layer and the fibrous layer is not demonstrated.

We performed RNAscope experiments to locate Pi16-expressing and *Postn*-expressing cells in the uninjured periosteum. We observed that Pi16-expressing cells are in the external fibrous layer of the periosteum while *Postn*-expressing cells are located along the cortex in the cambium layer. The data are added in Fig 4B and Fig. 4- Supplementary Figure 1 and mentioned in the result section “Pi16-expressing SSPCs were located within the fibrous layer, while *Postn*-expressing cells were found in the cambium layer and corresponded to Runx2-expressing osteogenic cells (Fig. 4 – Supplementary Figure 1A-C).”.

(5) Interpretation of transplantation experiments in Figure 5 is not straightforward, as the authors did not demonstrate the purity of Prx1Cre-GFP+SCA1+ cells and Prx1Cre-GFP+CD146- cells to pSSPCs and IIFCs, respectively. It is possible that these populations contain much broader cell types beyond SSPCs or IIFCs.

We agree with the reviewer that our methodology for cell transplantation required more justification and validation. We decided to use a transgenic mouse line to be able to trace the cells in vivo after grafting. *Prx1* marks limb mesenchyme during development and the *Prx1Cre* mouse model allows to label all SSPCs contributing to callus formation. Therefore, we used *Prx1Cre*, *R26mTmG* mice as donors for SSPCs and IIFCs isolation (Duchamp de Lageneste et al. 2018; Logan et al. 2002). *Prx1* does not mark immune and endothelial cells but can label pericytes and fibroblastic populations (Duchamp de Lageneste et al. 2018; Logan et al. 2002;

Julien et al. 2021). In the uninjured periosteum, Sca1 (*Ly6a*) is only expressed by SSPCs and endothelial cells (Fig 3-Supplementary figure 2, Fig 6-Supplementary figure 1). We sorted GFP+ Sca1+ cells from uninjured periosteum of *Prx1Cre*; *R26mTmG* mice to isolate only SSPCs and excluding endothelial cells and pericytes. For IIFCs, we isolated cells at day 3 post-fracture, as in our snRNAseq data, we detected IIFCs but no SSPCs, chondrocytes or osteoblasts at this stage of repair. To eliminate Prx1-derived pericytes, we sorted GFP+CD146- cells, as CD146 is specifically expressed by pericytes. We added Figure 6-supplementary Figure 1 to better illustrate the expression of *Prx1*, SCA1 (*Ly6a*) and CD146 (*Mcam*) in the uninjured and day 3 post-fracture datasets. We further demonstrate the purity of SSPCs and IIFCs isolation by qPCR on sorted GFP+ Sca1+ cells from uninjured periosteum and GFP+ CD146- cells from day 3 post-fracture periosteum and hematoma and confirmed the absence of contamination by other cell populations (Figure 6-Supplementary figure 1E). We made the following changes in the text: “To functionally validate the steps of pSSPC activation, we isolated SCA1+ GFP+ pSSPCs from *Prx1Cre*; *R26mTmG* mice, excluding endothelial cells, and grafted them at the fracture site of wild-type hosts” and “we isolated GFP+ CD146- from the fracture callus of *Prx1Cre*; *R26mTmG* mice at day 3 post fracture, that correspond to IIFCs without contamination by pericytes (CD146+ cells) (Fig. 6C, Figure 6 – Supplementary Fig.1).

Reviewer #2 (Public Review):

Summary:

The authors described cell type mapping was conducted for both WT and fracture types. Through this, unique cell populations specific to fracture conditions were identified. To determine these, the most undifferentiated cells were initially targeted using stemness-related markers and CytoTrace scoring. This led to the identification of SSPC differentiating into fibroblasts. It was observed that the fibroblast cell type significantly increased under fracture conditions, followed by subsequent increases in chondrocytes and osteoblasts.

Strengths:

This study presented the injury-induced fibrogenic cell (IIFC) as a characteristic cell type appearing in the bone regeneration process and proposed that the IIFC is a progenitor undergoing osteochondrogenic differentiation.

Weaknesses:

This study endeavored to elucidate the role of IIFC through snRNAseq analysis and in vivo observation. However, such validation alone is insufficient to confirm that IIFC is an osteochondrogenic progenitor, and additional data presentation is required.

As mentioned in the response to Reviewer 1, the differentiation trajectory of SSPCs presented in our study is supported by a combination of bioinformatic analyses and in vivo validation:

- snRNAseq allowed us to identify the different populations in the uninjured periosteum. In silico, in vitro and in vivo analyses altogether showed that Sca1+ cells are the SSPC population (Fig 2E-G).

- At day 3 post-fracture, we did not detect Sca1+ cells in the callus (Fig 4 – Supplementary figure 2). Instead, we observed the appearance of a new population, IIFCs. This population clustered along SSPCs and pseudotime analyses indicate that SSPCs can differentiate into IIFCs (Fig 5B). We confirmed the ability of Sca1+ SSPCs to form IIFCs, by grafting them in the fracture callus and assessing their fate at day 5 post-fracture (Fig 6B).

- In silico, we observed that IIFCs clustered along osteogenic and chondrogenic cells. The pseudotime trajectory suggests that IIFCs can differentiate into both lineages (Fig 5B-C). This

is coherent with the progressive expression of osteochondrogenic genes observed in IIFCs (Fig 5C, Fig 8A, C, E). In vivo, we observed the progressive expression of Runx2 and Sox9 by IIFCs undergoing differentiation (Fig 6A). We now show that IIFCs are not undergoing apoptosis, indicating that these cells further differentiate (Fig 7 – Supp 2). To functionally assess the osteochondrogenic potential of IIFCs, we used transplantation assay and showed that Prx1-GFP+ IIFCs from day 3 post-fracture form cartilage and bone when transplanted at the fracture site of wild-type mice (Fig 6C).

We would like to insist on the robustness of the bioinformatic analyses performed in our study. First, we used datasets from different time points post-fracture to capture the true temporal progression of cell populations in the fracture callus. We used a large combination of tools shown to be reliable in many studies (Julien et al. 2021; Matsushita et al. 2020; Debnath et al. 2018; Baccin et al. 2020; Junyue Cao et al. 2019; Zhong et al. 2020), and all tools converge in the same trajectory. To further show the relevance of pseudotime in our model, we illustrate the distribution of the cell populations by time point (Fig. 5D). We can observe a parallel between the time points and the pseudotime, reinforcing that the pseudotime trajectory reflects the timing of SSPC differentiation. Overall, the combined in silico, in vitro and in vivo analyses strongly support that Sca1+ Pi16+ cells are the periosteal SSPC population, specifically represented in the uninjured dataset. In response to bone fracture, these SSPCs give rise to IIFCs that are specifically represented in the intermediate stages (days 3 and 5) prior to osteochondrogenic differentiation.

We made the following changes in the text:

- Line 81-87: “We performed in vitro CFU assays with sorted GFP+SCA1+ and GFP+SCA1- cells isolated from the periosteum of Prx1Cre; R26mTmG mice, as Prx1 labels all SSPCs contributing to the callus formation1. Prx1-GFP+ SCA1+ showed increased CFU potential, confirming their stem/progenitor property (Fig 2F-G). Then, we grafted Prx1GFP+ SCA1+ et Prx1-GFP+ SCA1- periosteal cells at the fracture site of wild-type mice. Only SCA1+ cells formed cartilage and bone after fracture indicating that SCA1+ cells correspond to periosteal SSPCs with osteochondrogenic potential (Fig 2H).”

- Line 120-122: “We did not detect Pi16-expressing SPPCs, consistent with the absence of cells expressing SSPC markers in day 3 snRNAseq dataset compared to uninjured periosteum (Fig. 4 – Supplementary Figure 2).”

- Line 170-172: “Only a small subset of IIFCs undergo apoptosis, further supporting that IIFCs are maintained in the fracture environment giving rise to osteoblasts and chondrocytes (Fig. 7 – Supplementary Figure 2).”

- Line 277-278: “Following this unique fibrogenic step, IIFCs do not undergo cell death but undergo either osteogenesis or chondrogenesis”

- Line 281-283: “During bone repair, this initial fibrogenic process is an integral part of the SSPC differentiation process, and a transitional step prior to osteogenesis and chondrogenesis.”

Reviewer #3 (Public Review):

In this manuscript, the authors explored the transcriptional heterogeneity of the periosteum with single nuclei RNA sequencing. Without prior enrichment of specific populations, this dataset serves as an unbiased representation of the cellular components potentially relevant to bone regeneration. By describing single-cell cluster profiles, the authors characterized over 10 different populations in combined steady state and post-fracture periosteum, including stem cells (SSPC), fibroblast, osteoblast, chondrocyte, immune cells, and so on. Specifically, a developmental trajectory was computationally inferred using the continuum of gene expression to connect SSPC,

injury-induced fibrogenic cells (IIFC), chondrocyte, and osteoblast, showcasing the bipotentials of periosteal SSPCs during injury repair. Additional computational pipelines were performed to describe the possible gene regulatory network and the expected pathways involved in bone regeneration. Overall, the authors provided valuable insights into the cell state transitions during bone repair and proposed sets of genes with possible involvements in injury response.

While the highlights of the manuscript are the unbiased characterization of periosteal composition, and the trajectory of SSPC response in bone fracture response, many of the conclusions can be more strongly supported with additional clarifications or extensions of the analysis.

(1) As described in the method section, both the steady-state data and full dataset underwent integration before dimensional reduction and clustering. It would be appreciated if the authors could compare the post-integration landscapes of uninjured cells between steady state and full dataset analysis. Specifically, fibroblasts were shown in Figure 1C and 1E, and such annotations did not exist in Figure 2B. Will it be possible that the original 'fibroblasts' were part of the IIFC population?

As suggested, we now identified the fibroblast population from the uninjured periosteum in the integration of datasets from all time points (Figure 5B and Fig. 5 – Supplementary Figure 2). We identified 4 fibroblast populations in the uninjured periosteum: *Luzp2+*, *Cldn1+*, *Hsd11b1+* and *Csmd1+* fibroblasts. *Luzp2+* and *Cldn1+* fibroblasts are clustering distinctly from the other populations in the integrated dataset. *Hsd11b1+* fibroblasts blend with SSPCs and IIFCs in the integrated dataset probably due to the low cell number. Finally, *Csmd1+* fibroblasts are clustering at the interface between SSPCs and IIFCs likely because they correspond to differentiating cells both in the uninjured periosteum and in response to fracture. We modified the resolution of clustering in our subset dataset, in order to represent *Luzp2+* and *Cldn1+* fibroblasts as an isolated cluster (Figure 5B, cluster 10). In addition, both pseudotime (Fig. 5B) and gene regulatory network analyses (Fig. 7D), show that the fibroblast populations are distinct from the activation trajectory of SSPCs. We added the following sentence to the text “Fibroblasts from uninjured periosteum (*Hsd11b1+*, *Cldn1+* and *Luzp2+* cells corresponding to cluster 10 of Fig. 5B) clustered separately from the other populations, suggesting the absence of their contribution to bone healing.”

(2) According to Figure 2, immune cells were taking a significant abundance within the dataset, specifically during days 3 & 5 post-fracture. It will be interesting to see the potential roles that immune cells play during bone repair. For example, what are the biological annotations of the immune clusters (B, T, NK, myeloid cells)? Are there any inflammatory genes or related signals unregulated in these immune cells? Do they interact with SSPC or IIFC during the transition?

In this manuscript, we report the overall dataset and focused our analyses on the response of SSPCs to injury and their differentiation trajectories. We did not include detailed analyses of the immune cell populations, that are out of scope of this manuscript and are part of another study (Hachemi et al, biorxiv, 2024)

(3) The conclusion of Notch and Wnt signaling in IIFC transition was not sufficiently supported by the analysis presented in the manuscript, which was based on computational inferences. It will be great to add in references supporting these claims or provide experimental validations examining selected members of these pathways.

The role of Wnt and Notch in bone repair has been widely studied and both signaling pathways are known to be regulators of SSPCs differentiation (Lee et al. 2021; Matthews et al. 2014; Novak et al. 2020; Wang et al. 2016; Kraus et al. 2022; Dishowitz et al. 2012; Junjie Cao et

al. 2017; Matsushita et al. 2020; Steven Minear et al. 2010; Steve Minear et al. 2010; Kang et al. 2007; Komatsu et al. 2010). It was previously shown that Notch inactivation at early stages of repair leads to bone non-union while Notch inactivation in chondrocytes and osteoblasts does not significantly affect healing, confirming its role in SSPC differentiation before osteochondral commitment (Wang et al. 2016). Wnt was shown to be a critical driver of osteogenesis (Matsushita et al. 2020; Steven Minear et al. 2010; Steven Minear et al. 2010; Kang et al. 2007; Komatsu et al. 2010), as Wnt inhibition alters bone formation and Wnt overactivation increases bone formation (Pinzone et al. 2009; Balemans et Van Hul 2007). The role of Wnt is specific to osteogenic engagement as Wnt inhibition promotes chondrogenesis (Hsieh et al. 2023; C.-L. Wu et al. 2021; Ruscitto et al. 2023). A study by Lee et al. recently confirmed the successive activation and crosstalk of Notch and Wnt pathways during osteogenic differentiation of SSPCs during bone healing (Lee et al. 2021). They showed a peak of Notch activation at day 3 post-injury followed by a progressive decrease that parallels an increase of Wnt signaling inducing osteogenic differentiation. These studies correlate with the sequential activation of Notch and Wnt observed in our snRNAseq analyses. Our analyses now reveal how this sequential activation of Notch and Wnt relates to the fibrogenic and osteogenic phase of SSPC differentiation respectively. We clarified this in the discussion and added the references above to support our claims.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) The manuscript is well-written overall. However, the authors often oversimplify outcomes and overstate the results. Some of the statements (delineated below) need to be recalibrated to be in line with the presented data.

In addition to the suggested conclusions, we also toned down the following ones to avoid overstating our results :

Line 24: suggesting a crucial paracrine role of this transient IIFC population

Line 227: suggesting their central role in mediating cell interactions after fracture

line 243: IIFCs produce paracrine factors that can regulate SSPCs

- Line 77 (86): The authors should add "might" before "correspond to".

We provided new sets of data including CFU experiments and transplantation assay to reinforce our conclusion. We replaced “correspond to” by “encompass”

- Line 102: SSPCs are obviously not "absent" in day 3 snRNAseq (Figure 2d). The percentage dropped (only) 75%, according to Figure 2e, which is far from disappearance. Overall, immunohistochemical staining is often dichotomous with snRNAseq designations. The authors should more carefully describe the results.

We agree that this comment may not reflect the data shown as we observe a strong decrease in the percentage of cells in SSPC clusters, but still detect few cells in the SSPC clusters. However, when we looked at the presence of Sca1⁺ Pi16⁺ cells at different time points, we confirmed the absence of cells expressing SSPC signature genes (Sca1, Pi16, Cd34) at day 3 injury. Due to the clustering resolution of the combined integration, some cells in the SSPC clusters might not be Sca1⁺ Pi16⁺. We now show these results in Fig. 4 – Supplementary Figure 2. We changed the text accordingly (line 120): “We did not detect *Pi16*-expressing SPPCs, consistent with the absence of cells expressing SSPC markers in the day 3 snRNAseq dataset compared to uninjured periosteum (Fig. 4 – Supplementary Figure 2)”.

- Line 134: The authors need to clearly state that GFP+IIFCs were isolated based on *Prx1CreGFP+CD146-*. The authors did not clearly demonstrate the relationship between *POSTN+* cells and *CD146-* cells, which poses concerns about the interpretation of transplantation experiments.

As mentioned above in response to reviewer 1-public review, we have clarified and provided additional information on our strategy to isolate SSPCs and IIFCs. We used the *Prx1Cre*; *R26mTmG* mice to mark all SSPCs and their derivatives with the GFP reporter in order to trace these populations after cell grafting. In the uninjured periosteum, *Sca1* (*Ly6a*) is only expressed by SSPCs and endothelial cells. We sorted GFP+*Sca1*+ cells to exclude endothelial cells. For IIFCs, we isolated cells at day 3 post-fracture, as in our snRNAseq data, we detect IIFCs but no SSPCs, chondrocytes or osteoblasts at this time point. However, we also detected pericytes that can be *Prx1*-derived. To eliminate potential pericyte contamination, we sorted GFP+ *CD146-* cells, as *CD146* is specifically expressed by pericytes. We added Figure 6-supplementary Figure 1 to better illustrate the expression of *Prx1*, *SCA1* (*Ly6a*) and *CD146* (*Mcam*) in the uninjured and day 3 post-fracture datasets. We further demonstrate the purity of SSPCs and IIFCs isolation by qPCR on sorted GFP+ *Sca1*+ cells from uninjured periosteum and GFP+ *CD146-* cells from day 3 postfracture periosteum and hematoma and confirmed the absence of contamination by other cell populations (Figure 6-Supplementary figure 1E). We made the following changes in the text (line 153): “To functionally validate the steps of pSSPC activation, we isolated *SCA1*+ GFP+ pSSPCs from *Prx1Cre*; *R26mTmG* mice, excluding endothelial cells, and grafted them at the fracture site of wild-type hosts” and “we isolated GFP+ *CD146-* from the fracture callus of *Prx1Cre*; *R26mTmG* mice at day 3 post fracture, that correspond to IIFCs without contamination by pericytes (*CD146+* cells) (Fig. 6C, Figure 6 – Supplementary Fig.1).

- Line 211: It is obvious from Figure 8F that ligand expression was not "specific" to the IIFC phase.

The data only shows a slight enrichment of ligand score.

We corrected the text by “ligand expression was increased during the IIFC phase”.

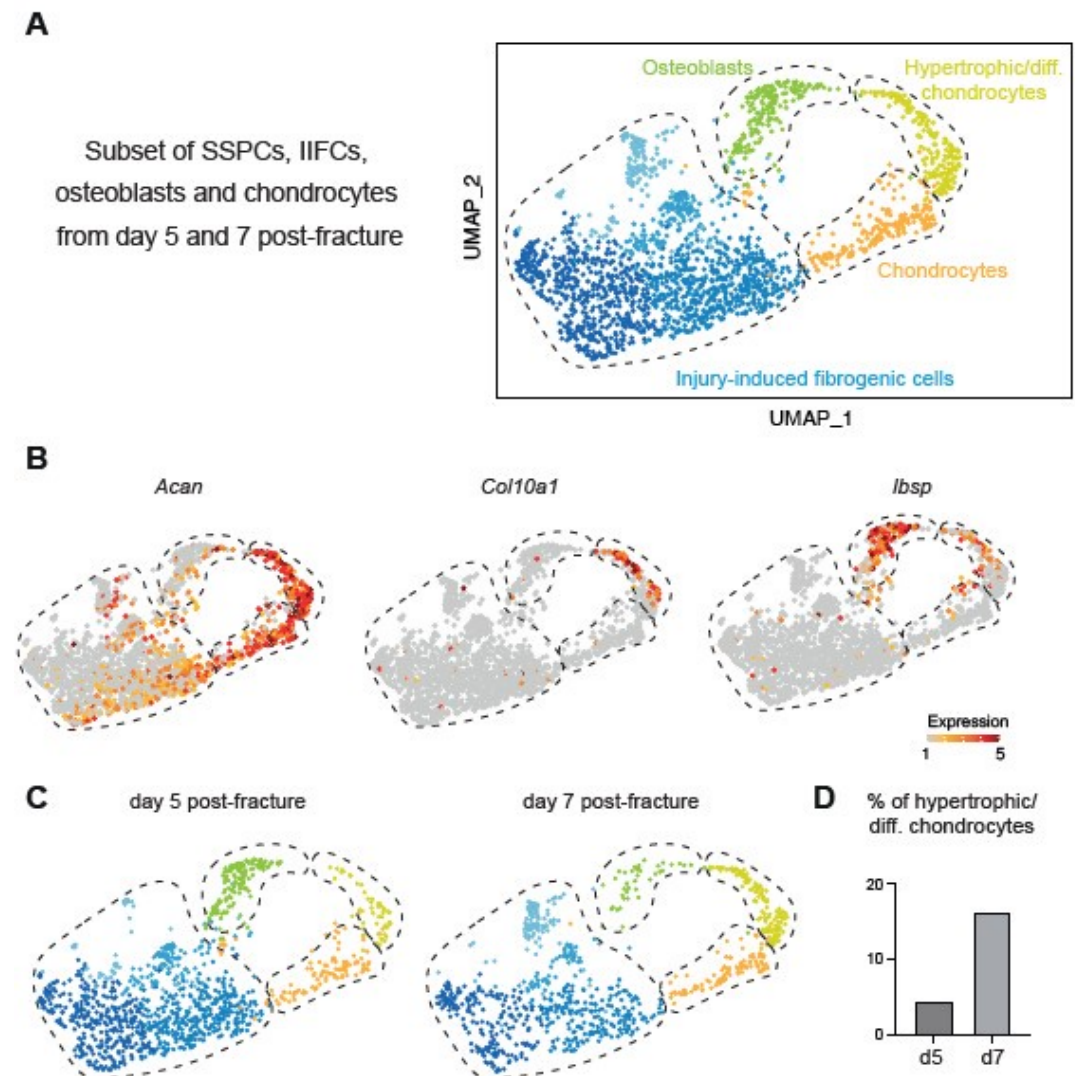
(2) Some of the computational predictions are incongruent with the known lineage trajectory. For example, *in vivo* lineage tracing experiments, including but not limited to, *PLoS Genet.* 2014. 10:e1004820, demonstrate that some of the chondrocytes within fracture callus can differentiate into osteoblasts. This is incompatible with the authors' conclusion that osteoblasts and chondrocytes represent two different terminal stages of cell differentiation in fracture healing. How do the authors reconcile this apparent inconsistency?

In this manuscript, we generated datasets corresponding to the initial stages of bone repair until day 7 post-injury. Therefore, our analyses encompass SSPC activation stages and engagement into osteogenesis and chondrogenesis. The results show that a portion of osteoblasts in the fracture callus are differentiating directly from IIFC via intramembranous ossification. The reviewer is correct to mention that osteoblasts have also been shown to derive from transdifferentiation of chondrocytes, which occurs at later stages of repair during the active phase of endochondral ossification (Julien et al. 2020; Aghajanian et Mohan 2018; Zhou et al. 2014; Hu et al. 2017). This process of chondrocyte to osteoblast transdifferentiation is not represented in our integrated dataset and may require adding later time points. However, when we analyzed the days 5 and 7 datasets independent of days 0 and 3, we were able to identify a cluster of hypertrophic chondrocytes (expressing *Col10a1*) connecting the clusters of chondrocytes and osteoblasts. This suggests that in this cluster,

hypertrophic chondrocytes are undergoing transdifferentiation into osteoblasts as shown in the Author response image 1. Additional time points are needed in a future study to perform in depth analyses of chondrocyte transdifferentiation.

Author response image 1:

Periosteum-derived chondrocytes undergo cartilage to bone transformation. A. UMAP projection of the subset of SSPCs, IIFCs, osteoblasts and chondrocytes in the integration of days 5 and 7 post-fracture datasets. B. Feature plots of *Acan*, *Col10a1* and *Ibsp* expression. C. UMAP projection separated by time points. D. Percentage of cells in the hypertrophic/differentiating chondrocyte cluster.



(3) The authors did not cite some of the studies that described the roles of Notch signaling in fracture healing, for example, *J Bone Miner Res.* 2014. 29:1283-94. The authors should test the specificity of Notch signaling activities to IIFCs (POSTN+ cells) in vivo.

The role of Notch in the activation of SSPCs during bone repair has been investigated in several studies (Lee et al. 2021; Matthews et al. 2014; Novak et al. 2020; Wang et al. 2016; Kraus et al. 2022; Dishowitz et al. 2012; Junjie Cao et al. 2017). Notch dynamic was previously

described with a peak at day 3 post-injury before a reduction when cells engage in osteogenesis and chondrogenesis (Lee et al. 2021; Dishowitz et al. 2012; Matthews et al. 2014). Notch plays a role in the early steps of SSPC activation prior to osteochondral differentiation as Notch inactivation in chondrocytes and osteoblasts does not affect bone repair (Wang et al. 2016). We added the references listed above to emphasize the correlation between our results and previous reports on the role of Notch and made changes in the discussion.

Reviewer #2 (Recommendations For The Authors):

Suggestions

(1) This research utilized snRNA seq for the basic hypothesis formation; however, the number of nuclei acquired was quite limited. Therefore, please explain the rationale for employing snRNA seq instead of scRNA seq, which includes cytoplasm, and additionally provide the markers used for cell type mapping in the scRNA analysis.

As mentioned in our response to reviewer #1 above, we analyzed a total of 6,213 nuclei from uninjured periosteum and fracture calluses at 3 stages of bone healing. We were able to describe 11 distinct cell populations including rare cell types in the fracture environment such Schwann cells, adipocytes and pericytes. The number of nuclei was sufficient to perform extensive analysis using a combination of cutting-edge algorithms. We agree that more nuclei would allow more indepth analyses of cell fate transitions and rare populations, such as pericytes and Schwann cells. However, we concentrated here on SSPC/fibrogenic cell that are well represented in our dataset. Our study robustness is also reinforced by the analysis of 4 successive time points to define the SSPC/fibrogenic cell trajectories. Our validations using immunohistochemistry and transplantation assays also confirmed that our dataset is sufficient to define cell trajectories. There is no clear consensus on the number of cells needed to perform scRNAseq analyses, as it depends on the cell types analyzed and the fold changes in gene expression. Previously reported single cell datasets containing a lower number of cells reached major conclusions including SSPC identification, cell differentiation trajectories and differential gene expression (658 cells in(Debnath et al. 2018), 300 in (Ambrosi et al. 2021) around 175 in(Remark et al. 2023))

Several studies have shown that snRNAseq provide data quality equivalent to scRNAseq in terms of cell type identification, number of detected genes and downstream analyses (Selewa et al. 2020; Wen et al. 2022; Ding et al. 2020; H. Wu et al. 2019; Machado et al. 2021). While, snRNAseq do not allow the detection of cytoplasm RNA, there is several advantages in using this technique:

(1) better representation of the cell types. To perform scRNAseq, a step of enzymatic digestion is needed. This usually leads to an overrepresentation of some cell types loosely attached to the ECM (immune cells, endothelial cells) and a reduced representation of cell types strongly attached to the ECM, such as chondrocytes and osteoblasts. In addition, large or multinucleated cells like hypertrophic chondrocytes and osteoclasts are too big to be sorted and encapsidated using 10X technology. Here, we optimized a protocol to mechanically isolate nuclei from dissected tissues that allows us to capture the diversity of cell types in periosteum and fracture callus.

(2) higher recovery of nuclei. We performed both isolation of cells and nuclei from periosteum in our study and observed that nuclei extraction is the most efficient way to isolate cells from the periosteum and the fracture callus.

(3) reduction of isolation time and cell stress. Previous studies showed that enzymatic digestion causes cell stress and induces stem cell activation (Machado et al. 2021; van den Brink et al. 2017). Therefore, we decided to perform snRNAseq to analyze the transcriptome of the intact periosteum without digestion induced-biais.

We added this sentence in the result section: “Single nuclei transcriptomics was shown to provide results equivalent to single cell transcriptomics, but with better cell type representation and reduced digestion-induced stress response (Selewa et al. 2020; Wen et al. 2022; Ding et al. 2020; H. Wu et al. 2019; Machado et al. 2021)”.

The list of genes used for cell type mapping are presented in Figure 3 – Supplementary figure 1. We added a detailed dot plot as Figure 3 – Supplementary figure 2.

(2) During the fracture healing process of long bones, the influx of fibroblasts is a relatively common occurrence, and the fibrous callus that forms during bone repair and regeneration is reported to disappear over time. Therefore, inferring that IIFC differentiates into osteo- and chondrogenic cells based solely on their simultaneous appearance in the same time and space is challenging. More detailed validation is necessary, beyond what is supported by bioinformatics analysis.

The first step of bone repair is the formation of a fibrous callus, before cartilage and bone formation. There are no data in the literature demonstrating that an influx of fibroblasts occurs at the fracture site. Several studies now show that cells involved in callus formation are recruited locally (i.e. from the bone marrow, the periosteum and the skeletal muscle surrounding the fracture site) (Duchamp de Lageneste et al. 2018; Julien et al. 2021; Colnot 2009; Jeffery et al. 2022; Debnath et al. 2018; Matsushita et al. 2020; Julien et al. 2022; Matthews et al. 2021). The contribution of locally activated SSPCs to the fibrous callus is less well understood. Lineage tracing shows that GFP⁺ cell populations traced in Prx1Cre-GFP mice include SSPCs, IIFCs, chondrocytes and osteoblasts.

The timing of the cell trajectories observed in our dataset correlates with the timing of callus formation previously described in the literature as the day 3 post-fracture mostly contains IIFCs while chondrocytes and osteoblasts appear from day 5 post-fracture. We conclude that IIFCs differentiate into osteochondrogenic cells based on multiple evidence beside the simultaneous appearance in time and space:

- In silico trajectory analyses identify a trajectory from SSPCs to osteochondrogenic cells via IIFCs. We added an analysis to show that our pseudotime trajectory parallels the timepoints of the dataset, confirming that the differentiation trajectory follows the timing of cell differentiation (Figure 5D).

- We show that IIFCs start to express chondrogenic and osteogenic genes prior to engaging into chondrogenesis and osteogenesis. In addition, we detected activation of osteo- and chondrogenic specific transcription factors in IIFCs. This shows a differentiation continuum between SSPCs, IIFCs, and osteochondrogenic cells (Figures 6-8).

- Using transplantation assay, we showed that IIFCs form cartilage and bone, therefore reinforcing the osteochondrogenic potential of this population (Figure 6B).

- IIFCs do not undergo apoptosis. We assessed the expression of apoptosis-related genes by IIFCs and did not detect expression. This was confirmed by cleaved caspase 3 immunostaining showing that a very low percentage of cells in the early fibrotic tissue undergo apoptosis.

Therefore, the idea that the initial fibrous callus is replaced by a new influx of SSPCs or committed progenitors is not supported by recent literature and is not observed in our dataset containing all cell types from the periosteum and fracture site. Overall, our bioinformatic analyses combined with our in vivo validation strongly support that IIFCs are differentiating into chondrocytes and osteoblasts during bone repair. Additional in vivo

functional studies will aim to further validate the trajectory and investigate the critical factors regulating this process.

(3) The influx of most osteogenic progenitors to the bone fracture site typically appears after postfracture day 7. It's essential to ascertain whether the osteogenic cells observed at the time of this study differentiated from IIFC or migrated from surrounding mesenchymal stem cells.

As mentioned above, there is not clear evidence in the literature indicating an influx of osteoprogenitors. Cells involved in callus formation are recruited locally and predominantly from the periosteum (Duchamp de Lageneste et al. 2018; Julien et al. 2021; Colnot 2009; Jeffery et al. 2022; Debnath et al. 2018; Matsushita et al. 2020; Matthews et al. 2021; Julien et al. 2022). Our datasets therefore include all cell populations that form the callus. Other sources of SSPCs include the surrounding muscle that contributes mostly to cartilage, and bone marrow that contributes to a low percentage of the callus osteoblasts in the medullary cavity (Julien et al. 2021; Jeffery et al. 2022). We provide evidence that IIFCs give rise to osteogenic cells using our bioinformatic analyses and in vivo transplantation assay (listed in the response above). As indicated in our response to reviewer #1, the steps leading to osteogenic differentiation observed in our dataset reflect the first step of callus ossification and correspond to the process of intramembranous ossification (up to day 7 post-injury). Endochondral ossification also contributes to osteoblasts including the transdifferentiation of chondrocytes into osteoblasts (Julien et al. 2020; Zhou et al. 2014; Hu et al. 2017). While this process mostly occurs around day 14 postfracture, we begin to detect this transition in our integrated day 5-day 7 dataset as shown in Author response image 1.

(4) It's crucial to determine whether the IIFC appearing at the fracture site contributes to the formation of the callus matrix or undergoes apoptosis during the fracture healing process. In the early steps of bone repair, the callus is mostly composed of an extracellular matrix (ECM). IIFCs are expressing high levels of ECM genes, including Postn, Aspn and collagens (Col3a1, Col5a1, Col8a1, Col12a1) (Figure 3 – Supplementary Figures 1-2 and Fig. 7 – Supplementary Figure 1B). IIFCs are the cells expressing the highest levels of matrix-related genes compared to the other cell types in the fracture environment (i.e. immune cells, endothelial cells, Schwann cells, pericytes, ...) as shown now in Fig. 7 – Supplementary Figure 1A. Therefore, IIFCs are the main contributors to the callus matrix.

We investigated if IIFCs undergo apoptosis. We observed that only a low percentage of IIFCs express apoptosis-related genes and are positive for cleaved caspase 3 immunostaining at days 3, 5 and 7 of bone repair. This shows that IIFCs do not undergo apoptosis and reinforces our model in which IIFCs further differentiate into osteoblasts and chondrocytes. We added these data in Fig. 7 – Supplementary Figure 2 and added the sentence in the results section “Only a small subset of IIFCs undergo apoptosis, further supporting that IIFCs are maintained in the fracture environment giving rise to osteoblasts and chondrocytes (Fig. 7 – Supplementary Figure 2).”

(5) Results from the snRNA seq highlight the paracrine role of IIFC, and verification is needed to ensure that the effect this has on surrounding osteogenic lineages is not misinterpreted.

To assess cell-cell interactions, we used tools such as Connectome and CellChat to infer and quantify intercellular communication networks between cell types. Studies showed the robustness of these tools combined with in vivo validation (Sinha et al. 2022; Alečković et al. 2022; Li et al. 2023). Here we used these tools to illustrate the paracrine profile of IIFCs, but in vivo validation would be required using gene inactivation to assess the requirement of

individual paracrine factors. We performed extensive analyses of the crosstalk between immune cells and SSPCs using our dataset in another study combined with in vivo validation, showing the robustness of the tool and the dataset (Hachemi et al. 2024). We adjusted our conclusions to reflect our analyses: “suggesting a crucial paracrine role of this transient IIFC population during fracture healing”, “suggesting their central role in mediating cell interactions after fracture”, “suggesting that SSPCs can receive signals from IIFC”.

References

- Aghajanian, Patrick, et Subburaman Mohan. 2018. “The Art of Building Bone: Emerging Role of Chondrocyte-to-Osteoblast Transdifferentiation in Endochondral Ossification”. *Bone Research* 6 (1): 19. <https://doi.org/10.1038/s41413-018-0021-z>.
- Alečković, Maša, Simona Cristea, Carlos R. Gil Del Alcazar, Pengze Yan, Lina Ding, Ethan D. Krop, Nicholas W. Harper, et al. 2022. “Breast Cancer Prevention by Short-Term Inhibition of TGFβ Signaling”. *Nature Communications* 13 (1): 7558. <https://doi.org/10.1038/s41467-02235043-5>.
- Ambrosi, Thomas H., Owen Marecic, Adrian McArdle, Rahul Sinha, Gunsagar S. Gulati, Xinming Tong, Yuting Wang, et al. 2021. “Aged Skeletal Stem Cells Generate an Inflammatory Degenerative Niche”. *Nature* 597 (7875): 256-62. <https://doi.org/10.1038/s41586-021-03795-7>.
- Baccin, Chiara, Jude Al-Sabah, Lars Velten, Patrick M. Helbling, Florian Grünschlager, Pablo Hernández-Malmierca, César Nombela-Arrieta, Lars M. Steinmetz, Andreas Trumpp, et Simon Haas. 2020. “Combined Single-Cell and Spatial Transcriptomics Reveal the Molecular, Cellular and Spatial Bone Marrow Niche Organization”. *Nature Cell Biology* 22 (1): 38-48. <https://doi.org/10.1038/s41556-019-0439-6>.
- Balemans, Wendy, et Wim Van Hul. 2007. “The Genetics of Low-Density Lipoprotein Receptor-Related Protein 5 in Bone: A Story of Extremes”. *Endocrinology* 148 (6): 2622-29. <https://doi.org/10.1210/en.2006-1352>.
- Brink, Susanne C van den, Fanny Sage, Ábel Vértesy, Bastiaan Spanjaard, Josi Peterson-Maduro, Chloé S Baron, Catherine Robin, et Alexander van Oudenaarden. 2017. “Single-Cell Sequencing Reveals Dissociation-Induced Gene Expression in Tissue Subpopulations”. *Nature Methods* 14 (10): 935-36. <https://doi.org/10.1038/nmeth.4437>.
- Cao, Junjie, Yalin Wei, Jing Lian, Lunyun Yang, Xiaoyan Zhang, Jiaying Xie, Qiang Liu, Jinyong Luo, Baicheng He, et Min Tang. 2017. “Notch Signaling Pathway Promotes Osteogenic Differentiation of Mesenchymal Stem Cells by Enhancing BMP9/Smad Signaling”. *International Journal of Molecular Medicine* 40 (2): 378-88. <https://doi.org/10.3892/ijmm.2017.3037>.
- Cao, Junyue, Malte Spielmann, Xiaojie Qiu, Xingfan Huang, Daniel M. Ibrahim, Andrew J. Hill, Fan Zhang, et al. 2019. “The Single-Cell Transcriptional Landscape of Mammalian Organogenesis”. *Nature* 566 (7745): 496-502. <https://doi.org/10.1038/s41586-019-0969-x>.
- Colnot, Céline. 2009. “Skeletal Cell Fate Decisions Within Periosteum and Bone Marrow During Bone Regeneration”. *Journal of Bone and Mineral Research* 24 (2): 274-82. <https://doi.org/10.1359/jbmr.081003>.
- Debnath, Shawon, Alisha R. Yallowitz, Jason McCormick, Sarfaraz Lalani, Tuo Zhang, Ren Xu, Na Li, et al. 2018. “Discovery of a Periosteal Stem Cell Mediating Intramembranous Bone Formation”. *Nature* 562 (7725): 133-39. <https://doi.org/10.1038/s41586-018-0554-8>.
- Ding, Jiarui, Xian Adiconis, Sean K. Simmons, Monika S. Kowalczyk, Cynthia C. Hession, Nemanja D. Marjanovic, Travis K. Hughes, et al. 2020. “Systematic Comparison of Single-Cell and Single-Nucleus RNA-Sequencing Methods”. *Nature Biotechnology* 38 (6): 737-46.

<https://doi.org/10.1038/s41587-020-0465-8>.

Dishowitz, Michael I., Shawn P. Terkhorn, Sandra A. Bostic, et Kurt D. Hankenson. 2012. “Notch Signaling Components Are Upregulated during Both Endochondral and Intramembranous Bone Regeneration”. *Journal of Orthopaedic Research* 30 (2): 296-303. <https://doi.org/10.1002/jor.21518>.

Duchamp de Lageneste, Oriane, Anaïs Julien, Rana Abou-Khalil, Giulia Frangi, Caroline Carvalho, Nicolas Cagnard, Corinne Cordier, Simon J. Conway, et Céline Colnot. 2018. “Periosteum Contains Skeletal Stem Cells with High Bone Regenerative Potential Controlled by Periostin”. *Nature Communications* 9 (1): 773. <https://doi.org/10.1038/s41467-018-03124-z>.

Hsieh, Chen-Chan, B. Linju Yen, Chia-Chi Chang, Pei-Ju Hsu, Yu-Wei Lee, Men-Luh Yen, ShawFang Yet, et Linyi Chen. 2023. “Wnt Antagonism without TGF β Induces Rapid MSC Chondrogenesis via Increasing AJ Interactions and Restricting Lineage Commitment”. *iScience* 26 (1): 105713. <https://doi.org/10.1016/j.isci.2022.105713>.

Hu, Diane P., Federico Ferro, Frank Yang, Aaron J. Taylor, Wenhan Chang, Theodore Miclau, Ralph S. Marcucio, et Chelsea S. Bahney. 2017. “Cartilage to Bone Transformation during Fracture Healing Is Coordinated by the Invading Vasculature and Induction of the Core Pluripotency Genes”. *Development* 144 (2): 221-34. <https://doi.org/10.1242/dev.130807>.

Jeffery, Elise C., Terry L.A. Mann, Jade A. Pool, Zhiyu Zhao, et Sean J. Morrison. 2022. “Bone Marrow and Periosteal Skeletal Stem/Progenitor Cells Make Distinct Contributions to Bone Maintenance and Repair”. *Cell Stem Cell* 29 (11): 1547-1561.e6. <https://doi.org/10.1016/j.stem.2022.10.002>.

Julien, Anais, Anuya Kanagalingam, Ester Martínez-Sarrà, Jérôme Megret, Marine Luka, Mickaël Ménager, Frédéric Relaix, et Céline Colnot. 2021. “Direct contribution of skeletal muscle mesenchymal progenitors to bone repair”. *Nature Communications* 12 (1): 2860. <https://doi.org/10.1038/s41467-021-22842-5>.

Julien, Anais, Simon Perrin, Oriane Duchamp de Lageneste, Caroline Carvalho, Morad Bensidhoum, Laurence Legeai-Mallet, et Céline Colnot. 2020. “FGFR3 in Periosteal Cells Drives Cartilage-to-Bone Transformation in Bone Repair”. *Stem Cell Reports* 15 (4): 955-67. <https://doi.org/10.1016/j.stemcr.2020.08.005>.

Julien, Anais, Simon Perrin, Ester Martínez-Sarrà, Anuya Kanagalingam, Caroline Carvalho, Marine Luka, Mickaël Ménager, et Céline Colnot. 2022. “Skeletal Stem/Progenitor Cells in Periosteum and Skeletal Muscle Share a Common Molecular Response to Bone Injury”. *Journal of Bone and Mineral Research*, juin, jbmr.4616. <https://doi.org/10.1002/jbmr.4616>.

Kang, Sona, Christina N. Bennett, Isabelle Gerin, Lauren A. Rapp, Kurt D. Hankenson, et Ormond A. MacDougald. 2007. “Wnt Signaling Stimulates Osteoblastogenesis of Mesenchymal Precursors by Suppressing CCAAT/Enhancer-Binding Protein α and Peroxisome Proliferator Activated Receptor γ ”. *Journal of Biological Chemistry* 282 (19): 14515-24. <https://doi.org/10.1074/jbc.M700030200>.

Komatsu, David E., Michelle N. Mary, Robert Jason Schroeder, Alex G. Robling, Charles H. Turner, et Stuart J. Warden. 2010. “Modulation of Wnt Signaling Influences Fracture Repair”. *Journal of Orthopaedic Research* 28 (7): 928-36. <https://doi.org/10.1002/jor.21078>.

Hachemi, Yasmine, Simon Perrin, Maria Ethel, Anais Julien, Julia Vettese, Blandine Geisler, Christian Göritz, et Céline Colnot. 2024. “Multimodal Analyses of Immune Cells during Bone Repair Identify Macrophages as a Therapeutic Target in Musculoskeletal Trauma”. <https://doi.org/10.1101/2024.04.29.591608>.

- Kraus, Jessica M., Dion Giovannone, Renata Rydzik, Jeremy L. Balsbaugh, Isaac L. Moss, Jennifer L. Schwedler, Julien Y. Bertrand, et al. 2022. "Notch Signaling Enhances Bone Regeneration in the Zebrafish Mandible". *Development* 149 (5): dev199995. <https://doi.org/10.1242/dev.199995>.
- Lee, S., L. H. Remark, A. M. Josephson, K. Leclerc, E. Muiños Lopez, D. J. Kirby, Devan Mehta, et al. 2021. "Notch-Wnt Signal Crosstalk Regulates Proliferation and Differentiation of Osteoprogenitor Cells during Intramembranous Bone Healing". *Npj Regenerative Medicine* 6 (1): 29. <https://doi.org/10.1038/s41536-021-00139-x>.
- Li, Jiaoduan, Dongyan Cao, Lixin Jiang, Yiwen Zheng, Siyuan Shao, Ai Zhuang, et Dongxi Xiang. 2023. "ITGB2-ICAM1 Axis Promotes Liver Metastasis in BAP1-Mutated Uveal Melanoma with Retained Hypoxia and ECM Signatures". *Cellular Oncology (Dordrecht)*, décembre. <https://doi.org/10.1007/s13402-023-00908-4>.
- Logan, Malcolm, James F. Martin, Andras Nagy, Corrinne Lobe, Eric N. Olson, et Clifford J. Tabin. 2002. "Expression of Cre Recombinase in the Developing Mouse Limb Bud Driven by a Prxl Enhancer". *Genesis* 33 (2): 77-80. <https://doi.org/10.1002/gene.10092>.
- Machado, Léo, Perla Geara, Jordi Camps, Matthieu Dos Santos, Fatima Teixeira-Clerc, Jens Van Herck, Hugo Varet, et al. 2021. "Tissue Damage Induces a Conserved Stress Response That Initiates Quiescent Muscle Stem Cell Activation". *Cell Stem Cell* 28 (6): 1125-1135.e7. <https://doi.org/10.1016/j.stem.2021.01.017>.
- Matsushita, Yuki, Mizuki Nagata, Kenneth M. Kozloff, Joshua D. Welch, Koji Mizuhashi, Nicha Tokavanich, Shawn A. Hallett, et al. 2020. "A Wnt-Mediated Transformation of the Bone Marrow Stromal Cell Identity Orchestrates Skeletal Regeneration". *Nature Communications* 11 (1): 332. <https://doi.org/10.1038/s41467-019-14029-w>.
- Matthews, Brya G, Danka Grcevic, Liping Wang, Yusuke Hagiwara, Hrvoje Roguljic, Pujan Joshi, Dong-Guk Shin, Douglas J Adams, et Ivo Kalajzic. 2014. "Analysis of αSMA-Labeled Progenitor Cell Commitment Identifies Notch Signaling as an Important Pathway in Fracture Healing". *Journal of Bone and Mineral Research* 29 (5): 1283-94. <https://doi.org/10.1002/jbmr.2140>.
- Matthews, Brya G, Sanja Novak, Francesca V Sbrana, Jessica L Funnell, Ye Cao, Emma J Buckels, Danka Grcevic, et Ivo Kalajzic. 2021. "Heterogeneity of Murine Periosteum Progenitors Involved in Fracture Healing". *eLife* 10 (février): e58534. <https://doi.org/10.7554/eLife.58534>.
- Minear, Steve, Philipp Leucht, Samara Miller, et Jill A Helms. 2010. "rBMP Represses Wnt Signaling and Influences Skeletal Progenitor Cell Fate Specification during Bone Repair". *Journal of Bone and Mineral Research* 25 (6): 1196-1207. <https://doi.org/10.1002/jbmr.29>.
- Minear, Steven, Philipp Leucht, Jie Jiang, Bo Liu, Ariel Zeng, Christophe Fuerer, Roel Nusse, et Jill A. Helms. 2010. "Wnt Proteins Promote Bone Regeneration". *Science Translational Medicine* 2 (29). <https://doi.org/10.1126/scitranslmed.3000231>.
- Novak, Sanja, Emilie Roeder, Benjamin P. Sinder, Douglas J. Adams, Chris W. Siebel, Danka Grcevic, Kurt D. Hankenson, Brya G. Matthews, et Ivo Kalajzic. 2020. "Modulation of Notch1 Signaling Regulates Bone Fracture Healing". *Journal of Orthopaedic Research* 38 (11): 2350-61. <https://doi.org/10.1002/jor.24650>.
- Pinzone, Joseph J., Brett M. Hall, Nanda K. Thudi, Martin Vonau, Ya-Wei Qiang, Thomas J. Rosol, et John D. Shaughnessy. 2009. "The Role of Dickkopf-1 in Bone Development, Homeostasis, and Disease". *Blood* 113 (3): 517-25. <https://doi.org/10.1182/blood-2008-03-145169>.

Remark, Lindsey H., Kevin Leclerc, Malissa Ramsukh, Ziyang Lin, Sooyeon Lee, Backialakshmi Dharmalingam, Lauren Gillinov, et al. 2023. “Loss of Notch Signaling in Skeletal Stem Cells Enhances Bone Formation with Aging”. *Bone Research* 11 (1): 50. <https://doi.org/10.1038/s41413-023-00283-8>.

Ruscitto, Angela, Peng Chen, Ikue Tosa, Ziyi Wang, Gan Zhou, Ingrid Safina, Ran Wei, et al. 2023. “Lgr5-Expressing Secretory Cells Form a Wnt Inhibitory Niche in Cartilage Critical for Chondrocyte Identity”. *Cell Stem Cell* 30 (9): 1179-1198.e7. <https://doi.org/10.1016/j.stem.2023.08.004>.

Selewa, Alan, Ryan Dohn, Heather Eckart, Stephanie Lozano, Bingqing Xie, Eric Gauchat, Reem Elorbany, et al. 2020. “Systematic Comparison of High-Throughput Single-Cell and SingleNucleus Transcriptomes during Cardiomyocyte Differentiation”. *Scientific Reports* 10 (1): 1535. <https://doi.org/10.1038/s41598-020-58327-6>.

Sinha, Sarthak, Holly D. Sparks, Elodie Labit, Hayley N. Robbins, Kevin Gowing, Arzina Jaffer, Eren Kutluerk, et al. 2022. “Fibroblast Inflammatory Priming Determines Regenerative versus Fibrotic Skin Repair in Reindeer”. *Cell* 185 (25): 4717-4736.e25. <https://doi.org/10.1016/j.cell.2022.11.004>.

Wang, Cuicui, Jason A. Inzana, Anthony J. Mirando, Yinshi Ren, Zhaoyang Liu, Jie Shen, Regis J. O’Keefe, Hani A. Awad, et Matthew J. Hilton. 2016. “NOTCH Signaling in Skeletal Progenitors Is Critical for Fracture Repair”. *The Journal of Clinical Investigation* 126 (4): 1471-81. <https://doi.org/10.1172/JCI80672>.

Wen, Fei, Xiaojie Tang, Lin Xu, et Haixia Qu. 2022. “Comparison of Single-nucleus and Single-cell Transcriptomes in Hepatocellular Carcinoma Tissue”. *Molecular Medicine Reports* 26 (5): 339. <https://doi.org/10.3892/mmr.2022.12855>.

Wu, Chia-Lung, Amanda Dicks, Nancy Steward, Ruhang Tang, Dakota B. Katz, Yun-Rak Choi, et Farshid Guilak. 2021. “Single Cell Transcriptomic Analysis of Human Pluripotent Stem Cell Chondrogenesis”. *Nature Communications* 12 (1): 362. <https://doi.org/10.1038/s41467-02020598-y>.

Wu, Haojia, Yuhei Kiritani, Erin L. Donnelly, et Benjamin D. Humphreys. 2019. “Advantages of Single-Nucleus over Single-Cell RNA Sequencing of Adult Kidney: Rare Cell Types and Novel Cell States Revealed in Fibrosis”. *Journal of the American Society of Nephrology* 30 (1): 23-32. <https://doi.org/10.1681/ASN.2018090912>.

Zhong, Leilei, Lutian Yao, Robert J. Tower, Yulong Wei, Zhen Miao, Jihwan Park, Rojesh Shrestha, et al. 2020. “Single Cell Transcriptomics Identifies a Unique Adipose Lineage Cell Population That Regulates Bone Marrow Environment”. *eLife* 9 (avril): e54695. <https://doi.org/10.7554/eLife.54695>.

Zhou, Xin, Klaus von der Mark, Stephen Henry, William Norton, Henry Adams, et Benoit de Crombrughe. 2014. “Chondrocytes Transdifferentiate into Osteoblasts in Endochondral Bone during Development, Postnatal Growth and Fracture Healing in Mice”. Édité par Matthew L. Warman. *PLoS Genetics* 10 (12): e1004820. <https://doi.org/10.1371/journal.pgen.1004820>.

<https://doi.org/10.7554/eLife.92519.2.sa0>